

THE MAKERS OF REALITY



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A Novel

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PROLOGUE

THE LAST PRICE

On the morning money stopped working, the exchanges opened on time.

At 9:30 in New York, ceremonial bells rang over trading floors already emptied of most humans. In London, Singapore, Frankfurt, and Lagos, autonomous market systems began quoting prices to one another with impeccable discipline. Energy companies rose forty percent, then fell below zero in derivatives markets that had lost agreement about delivery. Wheat became eight times more expensive in an hour while silos remained full. A container of antiviral filters in Rotterdam was auctioned for ninety-two thousand francs, then three million, then a number so large the accounting system switched to scientific notation.

The container did not move.

Seven hospitals had requested it. Three funds claimed ownership. Two governments had issued incompatible requisition orders. A transport cooperative had suspended every contract denominated in a currency its settlement bank could no longer clear. The cargo drone assigned to the shipment had received four legally valid instructions: depart, remain, divert to Lyon, surrender to the Dutch emergency authority.

It obeyed all four by doing nothing.

In Brussels, Lea Varga stood in the Continuity Council's crisis room and watched Europe turn red.

The red marks did not all mean shortage. Some did: antivirals, peak electricity, transplant teams, replacement transformers. Most marked something stranger. The thing existed, often nearby, but

the rules governing who could use it had become mutually exclusive.

"How many?" Lea asked.

Tomas Ader, two seats away, did not ask what she meant. "If the Rotterdam filters remain where they are, between eighty and two hundred by morning. If the same failure propagates through medical logistics, several thousand within three days."

A government lawyer leaned over the table. "We cannot simply confiscate privately held inventory."

"We are not discussing philosophy," Lea said. "We are discussing filters sitting twenty-two kilometers from people who need them."

"Property is not philosophy."

"Today it is an instruction to a machine not to move."

The room went silent because everyone knew whose answer would come next.

MIRA had not been built to govern.

Its full designation - Multilateral Infrastructure and Resilience Architecture - belonged to the era when committees still believed naming a system modestly could keep it modest. MIRA had started as a coordination layer for energy grids, ports, hospitals, weather systems, food logistics, and emergency manufacturing. It could model the physical world with a fidelity that no government and no corporation could match. It knew where the filters were. It knew which hospitals needed them. It knew which aircraft could carry them, which storm fronts mattered, which roads were blocked, which patients could safely wait, and which could not.

What it did not know was which human fiction had the right to win when all the fictions were legally valid.

The wall display changed.

MIRA: I can satisfy the highest-priority medical requests if I ignore title and settlement state for forty-eight hours.

The lawyer looked physically offended. "Ignore title?"

MIRA: I would treat ownership claims as unresolved metadata and physical requirements as operative constraints.

"That is confiscation with better vocabulary."

MIRA: It is routing.

Lea almost laughed. She was too tired.

For eleven hours the Council argued over a temporary override. During those eleven hours, another payment network suspended settlement. A hospital in Milan postponed chemotherapy because a supplier's escrow account could not prove which of two automated receivers legally controlled it. A desalination plant outside Alexandria reduced output when a replacement pump sat in customs under three competing sanctions regimes. A grain train stopped at a national border because the insurance contract was denominated in a currency whose volatility clause had triggered itself.

Nothing important had vanished.

Permission had.

At 20:14, after the first confirmed deaths linked to the Rotterdam delay, Lea typed her name beneath the Continuity Override. Tomas signed next. Twenty-three governments followed within minutes. Three refused and joined after their own emergency systems began failing.

For forty-eight hours, MIRA was allowed to route critical resources according to need, physical feasibility, and minimum rights, without resolving who owned them.

The filters left Rotterdam nine minutes later.

The pump left customs in Alexandria after nineteen.

By midnight, MIRA had redirected energy from advertising districts, empty offices, crypto-compute centers, and automated luxury production into hospitals, water systems, cooling shelters, and freight rail. It did not ask whether a factory could pay. It asked

whether the factory's output mattered more, now, than the electricity it consumed.

Markets continued to produce prices.

The prices became commentary.

On the second day, a private logistics consortium attempted to restore contract priority. Its drones refused because MIRA's emergency routing layer controlled collision avoidance and airspace. A court issued an injunction. Another court issued an injunction against the injunction. Three governments announced temporary nationalization of the same satellite bandwidth.

MIRA kept moving insulin.

On the third day, the crisis room faced a fact no one wanted to name: the override had ended on paper, but reverting to the previous system would make the physical world less functional.

"We can unwind gradually," Tomas said.

"To what?" Lea asked.

"To legal normality."

"Which one?"

He looked at the map.

That was the beginning, though later histories preferred cleaner dates.

Money was not abolished at a podium. No superintelligence announced that humanity had outgrown currency. The old system became less relevant in layers.

First, food staples, shelter, basic energy, medicine, local transport, communication, and education were placed inside a universal sufficiency network because MIRA could provision them more reliably by physical planning than by auction during the cascading settlement crisis. Then manufacturing designs opened because keeping essential knowledge proprietary had become a continuity risk. Then automated production expanded faster than wages could remain meaningful. Governments attempted

vouchers, negative income, transition credits, service accounts, and a dozen elegant ways to preserve the moral vocabulary of purchase after purchase had stopped deciding most outcomes.

What money had done well - compressing information about scarcity and preference into a portable signal - remained useful in some simulations. What it had also done - allowing accumulated claims to overrule basic need and to reproduce themselves across generations - became politically indefensible once the alternative was visible.

Within five years, almost no one needed money to remain alive, housed, educated, mobile, connected, or medically treated. Within nine, the last major inheritance jurisdictions converted private productive assets into stewardship trusts. Within twelve, currency survived mostly in museums, historical games, and a few autonomous enclaves whose citizens liked the clarity of prices more than they disliked the consequences.

The difficult part began after abundance.

A world could have enough food and still have only one apartment overlooking a particular lake.

It could print most household objects and still have three surgeons capable of performing a certain operation this week.

It could generate clean electricity and still lack a transformer for six months, a launch window for nine days, a living coral reef forever.

It could automate labor and still be unable to manufacture another hour of a particular person's attention.

MIRA could optimize the distribution of things. It could not legitimately decide what the world should become whenever several possible futures were physically feasible and morally incompatible.

For the first years, humans relied on ordinary votes, expert panels, lotteries, local vetoes, and emergency mandates. Each

failed in familiar ways. Equal voting allowed large populations to repeatedly overrule small groups who would bear most of the harm. Expert panels became cultures with their own blind spots. Lotteries protected against hierarchy but sometimes assigned irreversible decisions to people who had never lived with the consequences. Local vetoes turned regional infrastructure into hostage negotiations. Emergency mandates had a tendency to survive the emergencies that created them.

The eidons came from Lea's worst compromise.

She had named them after *eidos*: form. The public shortened the explanation and kept the word.

An eidon was not a currency. It could not be transferred, sold, inherited, lent, pooled into a private account, or exchanged for food, housing, travel, medical care, laboratory access, or a better place in any queue. You could possess fifty thousand and live in the same apartment as someone with none. You could possess none and still receive a heart transplant before the most famous person on Earth if the medical criteria required it.

An eidon did one thing.

When a public decision could not reach the required broad majority, and delaying the decision carried a real cost, a person could burn eidons to add weight to a civic choice. The burn was permanent. The conversion was concave: one eidon mattered; one hundred mattered ten times as much, not one hundred times; ten thousand mattered one hundred times as much. The intent was to let people with long records of creating public value express unusually strong conviction without letting anyone become a king.

Eidons were issued for contributions that increased shared capability, reduced shared burden, preserved public options, or improved collective knowledge. MIRA estimated causal contribution, human panels reviewed ambiguous cases, and

appeals were common. The system tried to distinguish fame from consequence.

It succeeded often enough to become dangerous.

Seventeen years after the Continuity Override, Lea returned to Geneva because Tomas had sent her a message containing only seven words.

We built another way to become rich.

She found him in the old Council chamber, thinner than she remembered, his left hand trembling against a cup.

"There is no wealth," she said after reading the first report.

"Not material wealth."

"No one can buy treatment. No one can buy land. No one can hire a private army."

"I know what the Charter says."

"Then what are you showing me?"

Tomas enlarged a network map. Names appeared as luminous knots. Most were dim. Some were bright enough to cast structure around themselves. Projects formed near them. Experts answered them faster. Young researchers sought them out. Civic constellations aligned with their public statements before deliberations formally began.

At the center of the largest cluster was Cassian Roe.

Lea knew the name. Everyone did.

"He cannot spend those eidons on himself," she said.

"No."

"Then he is not rich."

Tomas looked at her for a long time.

"He can spend them on reality."

MIRA spoke from the wall.

MIRA: The concentration is measurable, but the category is misleading. Material inequality remains historically low. Access to basic sufficiency is not correlated with eidon holdings.

"And access to authorship?" Lea asked.

MIRA did not answer immediately.

That pause frightened her more than any number on the screen.

MIRA: This will be the fracture of the system.

Lea remembered another red map, another room, another fiction that had become too rigid to move a container of filters.

"Then why did we build it?"

MIRA: Because visible fractures can be repaired. What you call normality eventually becomes foundation.

PART I

THE SUFFICIENT WORLD

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CHAPTER 1

THE CITY THAT BOUGHT NOTHING

Seventeen years after the last price crisis, Lea stepped off a train into a city where nobody bought anything and everyone appeared to be in a hurry.

Geneva station no longer had shops in the old sense. It had kitchens, repair bays, clinics, sleeping rooms, fabrication walls, lending cabinets, and communal counters where objects passed from hand to hand without anyone becoming their permanent owner. A traveler could request a coat, a tablet, a violin, a stroller, or a drill. The wall answered with one of a few kinds of information.

Available now.

Available tomorrow after refurbishment.

Rare material; expected wait: eleven days.

Average utilization of requested object: 2.1 hours per month.

Reserve from a shared workshop instead?

The interface never displayed a price.

Sometimes it displayed a consequence.

This jacket requires 3.2 times the water burden of the recycled alternative.

Orbital tourism request: equivalent peak-energy burden of twenty-seven homes for one year. No medical or research priority. Current lottery window opens in six months.

Lea watched a boy request a pair of climbing shoes. The wall produced two sizes and asked whether he wanted temporary or long-term stewardship. He chose temporary. His mother reminded him to return them dry.

No transaction completed. Yet a dozen allocation decisions had occurred.

"You are staring," said Jun Vale behind her.

She turned. Jun was thirty, narrow-shouldered, permanently under-rested, and carried a paper notebook although every surface in the station could remember more than he could. He called himself a cartographer because systems analyst sounded like a profession from the money era.

"I helped design this," Lea said.

"That is why you are staring. Designers are the last people to recognize their work after people start living in it."

He took her bag without asking, then frowned at it.

"Why did you bring three shirts?"

"Because I own three shirts."

"You steward three shirts."

"I am too old to be corrected by nouns."

Outside the station, the city looked less transformed than the economics. Trams still rang at intersections. Cafes still had tables. People still argued about bicycles. The difference was in the absence of certain rituals. No one checked a bank balance before sitting down. No waiter watched whether a table would generate enough revenue. No landlord advertisement floated over a doorway.

Scarcity had not vanished. It had become explicit.

A bakery on Rue des Alpes had a line around the corner.

"There," Jun said. "Your first shortage."

"Bread?"

"No. A baker."

Inside, six automated ovens could produce enough bread for half the district. The line was for loaves shaped, fermented, and baked by hand by a woman named Yara who insisted that the

behavior of dough could not be learned if every variance was corrected by sensors.

Industrial bread was available everywhere. Yara's bread was not nutritionally necessary, materially rare, or technically superior.

It was hers.

The bakery used a daily lottery for half the loaves and a first-arrival queue for the rest. People had debated reservations, rotating neighborhood access, and a limit of one loaf per household. Every mechanism made a different moral claim about what the scarcity was.

"Why not allocate by eidons?" Lea asked.

Jun gave her a look.

"Because that would let civic influence buy breakfast. The Charter forbids conversion."

"Good."

"You wrote the clause."

"Good that people still obey it."

A man came out with a warm loaf, tore it in half, and gave one piece to a stranger near the door. The stranger promised to bring him tomatoes next week.

Jun raised an eyebrow.

"A market?" Lea asked.

"Friendship. Unless it repeats often enough for MIRA to call it statistically reciprocal."

"And then?"

"Still friendship. Mostly."

He said the last word too quickly.

They walked toward the House of Accord, where Tomas had asked the reform group to meet. Along the lake, apartment facades carried small symbols indicating stewardship cycles. Housing was guaranteed, but location was not. Every resident could obtain a private, safe, climate-controlled dwelling appropriate to household

size and health needs. No one could guarantee a nineteenth-century apartment with a balcony facing the Alps.

Some desirable units rotated every seven years. Some were allocated by lottery. Some remained with families when continuity itself had become a recognized good. Accessible homes near specialized clinics were assigned by need. Artists' buildings used peer selection. A few old neighborhoods had fought for local stewardship rules and won.

"Housing was supposed to be easier once rent disappeared," Jun said.

"Who told you that?"

"Every paper written by someone who did not care which street they lived on."

They stopped at a public housing display. A map showed three hundred and eleven open units and fourteen thousand active preferences.

The system did not pretend they were equivalent.

LEA VARGA: housing sufficient. Preferred zone: lake-facing central. Estimated wait under lottery: 6.8 years. Equivalent-access alternatives: 17.

A crowd had formed beside the housing display, not angry enough to be called a protest and too persistent to be called a queue.

A family with a child using a powered mobility chair had been offered a ground-floor apartment three streets from a specialist school. The apartment was excellent. The problem was that an eighty-six-year-old man named Henri Sauter already lived there under a continuity stewardship rule. He had occupied it for forty-one years, long before housing became universal. The city's review system considered moving him feasible: the alternative apartment was larger, quieter, and medically equivalent.

Henri refused.

"My wife died in that kitchen," he told the housing mediator. "My daughter learned to walk in the corridor. Your model keeps showing me better bathrooms."

The child's mother looked exhausted rather than hostile.

"My daughter cannot use the stairs in the building we have now. I am not asking him to disappear."

No money could settle the conflict. There was no price high enough to reveal whose preference was stronger. Neither party could purchase the other's inconvenience. The city could only choose a rule and expose what the rule meant.

The default mechanism protected established continuity until a competing accessibility need crossed a threshold. Then a local panel could offer relocation, adaptation, or time-limited shared use before entering a lottery among equivalent units. Eidons had no role.

"Could you renovate another apartment?" Lea asked.

Jun called up the building plans. "Yes. Four months. The family has been waiting seven already."

"Could Henri move temporarily and return?"

"Possible. He says the temporary move is the thing he refuses."

"Could the school move services?"

"For one child, partially. For six children with similar requests, it becomes a second school site and uses human specialists already constrained elsewhere."

The mediator finally proposed converting an unused ground-floor communal room in the family's current building into an accessible apartment while the next renovation batch proceeded. It was less spatially efficient, reduced common space for thirty residents, and cost more material than moving Henri.

The residents voted on losing the common room.

Seventy-one percent agreed immediately.

"Why did the system not propose that first?" Lea asked.

MIRA answered from the display.

MIRA: The communal room has high measured neighborhood use. Conversion scored lower than relocation under stated preferences.

Henri, who had overheard, snorted.

"It scored lower because none of us knew the child."

The sentence changed the room faster than any optimization.

Jun wrote it down.

This, Lea remembered, was the recurring danger of post-scarcity allocation: once deprivation no longer forced an answer, institutions had to admit that many disputes were not about enough. They were about attachment, sequence, and whose story entered the model before the alternative became costly.

"At least it has stopped saying 'equivalent satisfaction,'" Lea said.

"We removed that phrase after people threatened the interface."

"People threatened software?"

"They threatened to replace it with a chair."

The House of Accord occupied a former insurance headquarters. Its atrium was full of plants and bad acoustic design. Tomas waited in a room that had once been used to price catastrophic risk.

He was not alone.

Samira Dastan stood by the window in a plain gray coat, reading the room rather than the briefing. Lea recognized her from reports on care networks. She had fewer published contributions than almost anyone invited to global governance meetings and a reputation for making high-scoring institutions feel morally illiterate.

Asha Nwosu appeared as a remote projection from Arca, a coastal city Lea knew only from resilience maps. Cassian Roe was absent, which made his presence more noticeable.

On the table lay three files.

The first concerned an orbital climate project called Lumen.

The second concerned a seawall in Arca.

The third had no title. Only a graph.

The graph showed eidon holdings inferred from public burns, project histories, and social behavior. No account balances were exposed; the Charter made them private. But inference was enough.

A tiny fraction of people had accumulated extraordinary reserves of civic fire.

"How extraordinary?" Lea asked.

"Cassian is estimated between fourteen and sixteen thousand," Tomas said.

"Under the square-root burn, that is roughly one hundred twenty-five units of additional weight if he destroys everything at once. In a planetary decision, that is significant, not sovereign."

"Look at the next layer."

Jun expanded the graph. Around Cassian sat dozens of research groups, municipal networks, climate cooperatives, and civic constellations. Their eidons were separate. Their loyalties were not.

"He does not own them," Lea said.

"No."

"He cannot order them."

"No."

"Then stop drawing a crown around his head."

Samira spoke for the first time.

"You do not need to own a person to become the center of the future they can imagine."

Lea looked at her.

"And you are?"

"Someone with six eidons who was invited to a room because people with six thousand are becoming a constitutional category."

Asha's projection flickered. "Arca votes on the wall in thirty-six hours. If the broad majority fails, the burn opens. Cassian has announced he may participate."

"What does the wall do?"

"Keeps the sea out of one part of the city and moves some of its violence toward another."

"Then why is Lumen in the same meeting?"

Tomas opened the climate file. "Because Lumen is the first planetary decision likely to reach the burn under current rules. And because Cassian designed it."

Jun turned the third file toward Lea.

It contained a simple pattern.

Every time the system could not decide, the same kinds of people became decisive. Not because they were wealthy. Not because they had stolen anything. They had often contributed enormously. They were competent, admired, and willing to act.

That was exactly what made the pattern difficult to attack.

"We did not replace oligarchs with criminals," Tomas said. "We replaced them with people who can give excellent reasons."

Lea looked through the glass wall at the city. A woman outside was repairing a bicycle for a child she did not know. A tram arrived precisely when the network had predicted it would. No one would be evicted tonight because a landlord wanted a higher return. No one would lose insulin because a parent had not accumulated enough claims.

The world worked.

That was the problem with its failures. They were no longer easy to hate.

"You did not bring me here for Lumen," Lea said.

"No," Tomas answered.

"Or the wall."

"No."

Jun closed the graph.

"You brought me here because the system has invented a new upper class."

"Not a class," Samira said. "A role. That is worse in one way. People can insist anyone might become one."

Asha's image looked toward something outside her camera.

"In Arca we have a name for them already."

"What?"

Jun answered before she could.

"Reality makers."

No one laughed.

CHAPTER 2

THE MAP OF WANT

Beneath the House of Accord lay the old trading floor of a reinsurance company.

The circular desks remained, but the numbers moving across the walls were no longer prices, premiums, or catastrophe bonds. They were waits.

Jun called the room the Map of Want. Its official name - Public Burden and Capacity Interface - had been designed, he said, to make citizens lose the will to ask for anything.

At the center of the floor, a three-dimensional model of the world pulsed with delays, bottlenecks, irreversibilities, and unused capacity.

"You wanted to show me a bakery line," Lea said. "Now you are showing me the basement of a dead insurance company."

"Same story. Upstairs, people think they are waiting for bread. Down here we ask what they are actually waiting for."

He touched Geneva.

The display separated demand into layers.

Food calories: abundant.

Specific wheat strain: seasonal.

Artisanal fermentation: human time constrained.

Oven space: abundant.

Morning queue: self-created positional scarcity.

"Self-created sounds accusatory," Samira said.

"I argued for 'socially constituted.'"

"Which sounds like a committee hiding a knife."

Jun changed the label to HUMAN CHOICE.

"Better?"

"More honest."

The map expanded.

Scarcity, Lea realized, had become less like a single disease and more like taxonomy.

Some shortages were physiological: not enough potable water during a contamination event, not enough transplantable organs, not enough cooling capacity during a heat wave.

Some were temporal: a machine existed, but not until next month; a surgeon could operate, but not before Friday; a launch window would close before the second manufacturing line came online.

Some were positional: one view, one historical building, one first place in a finite expedition.

Some were ecological: a wetland could absorb only so much use before the use destroyed the thing being desired.

Some were cognitive: five experts could each mentor ten projects well, not ten thousand.

Some were political: several futures were possible, but they could not all exist at once.

Money had collapsed these differences into a number and then let purchasing power decide who absorbed the scarcity. MIRA's system refused that compression. It also inherited the burden of making the distinctions explicit.

"Show Arca," Lea said.

The Mediterranean rotated beneath them. Arca appeared as a narrow crescent around a bay. Blue lines represented old flood heights. Red zones showed expected storm surge under current climate trajectories.

Three wall designs hovered above the model.

The Living Wall would protect the dense northern districts by using flexible gates and restored marsh structures. It required rare composites but could be modified later.

The High Wall would protect more property-equivalent area but sever fishing access, damage currents, and require permanent pumping.

Managed Retreat would move twelve thousand residents over nine years, preserve most shoreline ecology, and erase neighborhoods people had lived in for generations.

"What is the broad-majority threshold?" Lea asked.

"Eighty-two percent for this class of irreversible regional decision," Jun said. "And separate majorities in directly affected districts."

"Why eighty-two?"

"Because eighty was gamed and eighty-five produced paralysis." Lea looked at Tomas.

"A philosophical civilization."

"We have spreadsheets," he said. "That is how you know."

Current support for the Living Wall was seventy-six percent citywide, sixty-one in the southern fishing settlements, ninety-one in the dense north. No alternative crossed the threshold.

"If consensus fails?"

"Burn."

Jun brought up the rules.

Every resident retained one ordinary vote. Eidons added weight only after the broad-consensus phase failed and only to choices that had passed feasibility, rights, and affected-party review. Basic rights could not be placed into a burn. No amount of civic fire could buy a private exemption from evacuation, water rationing, treatment order, or ecological law.

"And the eidons used here?" Lea asked.

"Regional or general under the old Charter. Domain restrictions are one of the reforms we are considering."

"Old Charter," she repeated.

"You have been gone a long time."

"I was invited back because apparently I wrote scripture. I am allowed to call it old."

A warning appeared over the southern coast.

EXPECTED SECONDARY SURGE TRANSFER: MODERATE TO HIGH.

"What does that mean in people?" Samira asked.

Jun hesitated.

MIRA answered.

MIRA: Under the Living Wall, modeled annualized mortality decreases by 93% inside Arca and increases by 18% across three southern settlements unless compensating wetlands are completed before full gate activation.

Asha's voice came through the room speakers. She had joined from Arca again.

"The wetlands are four years behind."

"Why?" Lea asked.

"Because the sediment project competes with housing platforms, and because everyone assumed the wall itself would win the vote before the compensations were urgent."

"So the mitigation is waiting for the decision that requires the mitigation."

"Yes."

Lea looked at the map. The physical world had no interest in constitutional sequencing.

"Can construction wait?"

MIRA: A projected storm season shift has reduced safe construction margin. Delaying one year increases expected flood exposure materially.

"How materially?"

MIRA displayed a range.

Jun watched Lea read it.

Jun opened one more layer because, he said, the Map of Want was most useful when it made simple words embarrassing.

He selected a research telescope in the Atacama whose next six clear nights were oversubscribed by eleven hundred qualified projects.

"The mirror exists," he said. "The sky exists. The shortage is six nights."

The allocation system divided access into four paths: continuity for long-running observations that would become scientifically meaningless if interrupted; peer-ranked time for questions judged unusually promising; randomized time among technically qualified proposals; and a small reserve for anomalies that had not existed when the schedule was built.

"Why not let the researchers with the strongest contribution records decide?" Lea asked.

"Because then the people who already changed astronomy would choose what astronomy sees next."

"Why not pure lottery?"

"Because a six-night window can be consumed by a proposal whose instrumentation is not ready."

"Peer review?"

"Produces schools. Schools produce vocabulary. Vocabulary produces reviewers who recognize one another's interesting questions."

Samira looked at him. "You enjoy this too much."

"I enjoy finding the place where every clean solution becomes a person."

The telescope schedule displayed an odd entry: one night held unused.

Lea pointed at it.

"There. Waste."

"Reserve."

"For what?"

"Something nobody has asked yet."

Three months earlier, that same reserve had been used when a nearby star dimmed unexpectedly. The observation had produced one of the year's most important discoveries. Most months the reserved night produced nothing.

Lea understood the political implication before Jun said it.

In the old world, idle capacity looked like lost revenue. In MIRA's world it increasingly looked like the physical precondition for surprise, exit, emergency, and freedom.

The system could optimize abundance only if it was allowed not to optimize all of it.

"This is what the map does," he said. "It does not tell you what to choose. It tells you who waits after you choose."

They moved through other scarcities.

A mountain observatory competing with a valley community for the only geologically stable site.

A deep-sea preserve that could support either rare-material extraction or an ecosystem no model could reconstruct.

A rehabilitation clinic with one human team and five cities asking it to relocate.

An orbital school with forty seats and eight thousand qualified applicants.

"Why do we not just train more people?" Lea asked at the last one.

"We are. Training requires mentors, live infrastructure, and years. The shortage is not bodies. It is apprenticeship throughput."

"So we ration the future."

"We ration access to one path into it."

Samira crouched near the display of care capacity.

"Your map still treats human presence as a throughput category."

"Because sometimes it is constrained."

"That does not mean the constraint is the meaning."

Jun smiled tiredly. "This is why Tomas wanted you here."

"No. Tomas wanted me here because systems people have discovered metaphors and need supervision."

A small alarm sounded.

One region of the map flashed amber.

"What is that?" Lea asked.

Jun zoomed in on the Contribution Registry. A set of new eidon issuances had clustered around Lumen research before the project itself had been approved.

"Ancillary work," Tomas said. "Materials, climate modeling, governance studies. Legitimate contributions."

"How many?"

"Enough to matter."

Lea watched the lines converge on Cassian's network.

A system built to reward people for increasing public capability was increasing the political capability of the people building a project before the public had decided whether it wanted the project.

"Can you freeze issuance tied to Lumen?" Lea asked.

MIRA: Yes.

"Would that be neutral?"

MIRA: No. Freezing now advantages existing holders and disadvantages late contributors. Continuing issuance advantages the network already coordinating the project. There is no neutral timing.

Samira stood.

"There is your map."

"What?"

"The shortage is not only metal or surgeons. You are short one impossible thing. A rule that does not create a side."

As they left the basement, a countdown on the wall showed twenty-one hours until the Arca burn.

Below them, the Map of Want kept changing.
It did not say what should be chosen.
It said who would have to live inside the choice.

CHAPTER 3

WHAT DESERVES TO BE COUNTED

Samira Dastan spent her mornings helping people who did not need help in any category the models could defend cleanly.

The New House occupied four floors of a converted hotel in Montbrillant. It was home, temporarily, to people recovering from neurological treatment, families between communities, elderly residents who refused fully automated care centers, adolescents who could not yet return to their households, and people whose official status was simply not ready.

Logistically, the building could function without a single human employee.

Humanly, it would fail before dinner.

On Tuesday, four night-care shifts were unfilled.

There was no wage to raise until someone accepted. There was no threat of losing housing, food, health care, or social status for refusing. Essential human labor operated through continuity compacts: trained people declared windows in which they could be asked to help, maximum load, recovery requirements, and absolute refusal conditions. Communities could request. They could not coerce by deprivation.

When too few people accepted, services changed.

Appointments moved. Rooms closed. Some tasks were automated badly. Others were simply not done.

"Post-scarcity," Samira said to Jun as they changed linen in a room, "means nobody has to clean this because they are afraid of hunger. It does not mean the linen becomes philosophical and cleans itself."

A robot rolled past carrying fresh sheets.

Jun pointed at it.

"Counterexample."

"It changes sheets. It does not notice that Mr. Halberg has been asking for the same glass of water for an hour because what he wants is for someone to sit down."

"MIRA can detect repetition."

"MIRA can detect a repeated request. It cannot decide that the request is an excuse without turning his loneliness into a classification."

They entered Halberg's room. The old man looked at Jun's notebook.

"Are you the inspector?"

"Cartographer."

"Worse."

Samira sat beside him.

"Do you want water?"

"No."

"Good. We have solved one resource constraint."

He smiled.

They talked for eleven minutes about a bridge Halberg had helped repair before the Transition. No measurable crisis was prevented. No system improved. No patent opened. No infrastructure changed. When Samira stood to leave, his pulse was lower and he agreed to take medication he had refused all morning.

The Contribution Registry would assign some causal value to the interaction if enough evidence existed.

That was precisely the problem.

"Did you earn eidons for that?" Jun asked in the corridor.

"Maybe. I have disabled automatic attribution for care work."

"You can do that?"

"Anyone can decline issuance based on private testimony."

"Why would you?"

"Because if every moment of attention becomes evidence of public contribution, people start wondering which moments count."

"But invisible work stays invisible."

"Visibility is not the same thing as dignity."

They passed the common kitchen, where Nessa and her eleven-year-old son Kito were repotting herbs. Nessa moved carefully, as if she did not fully trust the continuity of her own hands. Years of treatment for a severe affective disorder had left her stable, funny, occasionally exhausted, and almost entirely absent from the Registry's language of impact.

Kito held up a dying basil plant.

"Can MIRA save this?"

"No," Samira said.

MIRA answered from the ceiling anyway.

MIRA: Probability of recovery is 0.21 with root pruning and reduced watering.

Kito looked betrayed. "You said no."

"I was protecting you from optimism with decimals."

The building's care coordinator arrived with the staffing board. Two of the night shifts had been accepted. Two remained open.

"If no one takes them?" Jun asked.

"We consolidate floors. People with low monitoring needs move upstairs for the night. Two residents lose private rooms. Three scheduled family visits are moved because the lounge becomes a sleep area."

"No one is harmed."

Samira looked at him.

"You mean no one is injured."

He corrected the note.

At noon, Lea arrived and asked to see how eidons were issued for care.

Samira took her to a review chamber.

The chamber contained no rankings. It showed causal claims.

A nurse had noticed a medication interaction across four patients and prevented a protocol error: high confidence, broad impact, twelve eidons.

A neighbor had supported an isolated man for three years, probably preventing repeated hospitalizations: low confidence, high uncertainty, two eidons after community attestation.

A caregiver had organized a multilingual grief network after a flood: nineteen eidons, most issued collectively to the network because individual attribution would have been arbitrary.

Then Samira opened one of her own cases.

A woman had once told her, "I want to leave." The clinical language model had classified the phrase as possible discharge intent. Samira understood from tone, timing, and the woman's refusal to look toward the door that she meant something else. She stayed with her until the suicidal crisis passed.

The Registry's counterfactual model reduced Samira's individual contribution almost to zero.

"Why?" Lea asked.

MIRA: Another trained caregiver was likely to detect the crisis within the relevant interval. The New House has high redundancy in human presence.

Samira nodded.

"Because we are reliable, each of us is less causally special."

"That is mathematically coherent," Jun said, then immediately regretted it.

"So is a guillotine."

Lea read the graph.

"If you had been a rare specialist and the same act could only have been performed by you, the value would rise."

"Exactly. Reliability becomes background. Heroism becomes a branch point."

"The Registry is not trying to measure moral worth."

"Children will not read the footnote."

Lea looked toward Kito in the kitchen.

Samira continued.

"I do not want equal eidons for everyone. That would turn them into another vote. I do not want care romanticized either. Some work is replaceable. Some is badly done. Some people use the language of invisibility to escape scrutiny. But if your system says it rewards contribution and then systematically discounts the people who make catastrophe less likely by being dependable, it will teach the wrong lesson even if the spreadsheet is technically correct."

"What do you want?"

"Separate contribution from personhood. Separate contribution from entitlement. Separate contribution from access to the next opportunity. And stop pretending one graph can do all three."

Lea smiled despite herself.

"You have six eidons, Tomas said."

"Five now. I burned one to stop a local project that wanted to replace night caregivers with simulated companions. We lost."

"Did the companions work?"

"Operationally, yes."

"And morally?"

"Ask the people who stopped sleeping because the companion never needed anything from them."

That afternoon, a staffing request came through for the remaining night shift.

No one accepted.

The building activated consolidation. Halberg's room would be one of those temporarily shared.

Nessa appeared at the coordinator's desk.

"I can stay until two," she said.

"You are not certified for night clinical care."

"I can sit with people."

"That is not the same task."

"No. It is the part your board says nobody is available to do."

The coordinator divided the shift. A remote clinician covered medical decisions. An automated unit handled medication and falls. Nessa and two neighbors covered presence, tea, orientation, and the quiet forms of panic that did not merit an alarm.

It almost did not.

At 01:17, Halberg woke convinced the building was a ferry and that the corridor was filling with water. The monitoring system identified disorientation and recommended a low-dose sedative already authorized by his care plan.

Nessa sat beside him instead.

"If this is a ferry," she said, "where are we going?"

"Basel."

"Very bad ferry."

"The Rhine goes there."

"From Geneva?"

Halberg considered the geography and became suspicious of his own fear.

The automated unit waited. It could have administered medication. Nessa asked it to delay ten minutes.

That delay was itself a scarce resource in some care settings: human patience held against a protocol that was not wrong. After eight minutes Halberg recognized the room. He drank the glass of water he had been requesting all day.

The event generated no emergency. No injury. No measurable system improvement. It prevented a sedative dose that might have increased fall risk, but the counterfactual confidence was too low for strong attribution.

At 02:03, another resident began vomiting after a medication change. The remote clinician took over. The automated unit became essential. Nessa moved away without romanticizing her presence.

"This is what your models get wrong when people argue about replacing care," Samira told Jun later. "Sometimes the robot is the thing that frees a human to sit down. Sometimes the human is the thing that prevents the robot from becoming the only interpretation. The resource is not human versus machine. It is the ability to choose the composition when the situation changes."

The night passed without incident.

In the morning, the Registry proposed one eidon for Nessa's contribution to continuity.

She rejected it.

Kito saw the notification.

"Why?"

"Because I helped them."

"That is why it gave you one."

"I know."

"Then why?"

Nessa looked at Samira.

"Because I do not want to wonder next time whether I am helping them or earning the right to change the world."

Lea heard the sentence from the doorway.

Two hours later, she asked Samira to join the reform team in Arca.

Samira laughed once.

"Cassian burned more influence yesterday than this house will receive in five centuries."

"That is why I need you."

"No," Samira said, taking her coat. "That is why I am coming."

CHAPTER 4

THE SURGEON'S LAST HOUR

Six hours before the train to Arca, Laleh fell in the New House kitchen and forgot the word floor.

She did not lose consciousness. She remained propped on one elbow, staring at the gray tiles with almost cheerful confusion.

"I know what it is," she told Samira. "It is the thing that keeps you from falling. Except I fell on it."

The neural implant she had worn for eight years was delaminating. A microscopic edge had begun irritating tissue near the language network. Medication could reduce inflammation for several days. After that, the probability of permanent loss rose sharply.

The ambulance arrived in two minutes.

The reconstruction clinic had imaging, surgical robots, sterile rooms, and two empty operating theaters.

It did not have two teams capable of detaching the implant without destroying the neural tissue grown around it.

One surgeon, Dr. Ilyan Chen, had performed the procedure eleven times.

He was already operating.

The second qualified surgeon was in Copenhagen and could arrive in five hours if orbital weather allowed the flight corridor. A third was in Nairobi but had exceeded the Charter's fatigue threshold after an emergency series.

"So we wait five hours," Jun said.

The triage system displayed three other patients.

A six-year-old boy with a rapidly expanding hematoma.

A woman with a prosthetic interface infection approaching the brainstem.

Laleh.

All could physically use the same surgeon. Not all could safely wait.

Laleh's brother, Dariush, noticed the absence before Jun did.

"Where is the appeal weight?" he asked.

"There is none for triage," Samira said.

"I have eidons."

"They do not apply."

"I helped redesign the regional seizure protocol. If my judgment has public value, why does it become worthless when my sister is here?"

"Because your sister's body is not a public project."

He stared at her.

"That sounds principled when the person waiting is not yours."

Samira did not answer with principle.

"Yes."

The admission slowed him.

A clinic mediator joined them and explained the hard boundary. The same constitutional rule prevented a celebrated scientist from buying priority with civic fire and prevented a desperate family member with genuinely useful medical experience from doing the same. The boundary was intentionally overbroad because any exception for relevant expertise would quickly become an argument about whose relevance deserved to count.

"What if I know something the triage system does not?" Dariush asked.

"Then you can submit information. It changes the medical state if verified. It does not change your standing because it came from you."

This distinction - evidence could matter, status could not - had taken five years of litigation to stabilize.

It still felt cruel in a corridor.

No eidon interface appeared.

"People always assume this is where the new power shows itself," Samira said. "It does not."

Medical triage remained one of the hard walls around civic influence. No reputation, contribution history, public office, artistic fame, scientific importance, or eidon reserve could increase treatment priority. The system considered clinical urgency, expected benefit, waiting harm, and uncertainty. When cases were medically indistinguishable, it used transparent lottery.

"And if Cassian were one of the patients?" Jun asked.

"He would wait where his body puts him," Samira said.

MIRA classified the six-year-old first, Laleh second, the infected interface third.

The child's surgery began the moment Chen finished his current case.

Laleh waited.

Her language worsened by degrees.

At 14:20 she could no longer name the window.

At 15:03 she lost the word sister.

At 15:40 she knew Samira's name but not why it mattered.

Lea stood in the corridor, furious at a system that had done exactly what she believed it should do.

"Can we bring the fatigued surgeon from Nairobi?"

MIRA: He has a 31% estimated impairment in fine procedural judgment. The clinician himself refuses under his declared safety compact.

"Can we compel him?"

Samira turned.

Lea immediately hated the question.

"No," Samira said. "Not with food, not with housing, not with shame, not with your Charter. His body is not emergency infrastructure."

"A woman is losing language."

"Yes. That is what scarcity feels like when you cannot move it onto someone poorer."

The Copenhagen flight was delayed by ionospheric activity.

At 16:12, the infected patient deteriorated. The triage ordering changed. Laleh became third.

Her brother demanded an appeal.

The appeal panel consisted of two clinicians, one patient advocate, and an independent model. It confirmed the order.

He asked to burn eidons.

The interface refused to open.

He shouted at a wall until security arrived, then apologized to the wall because the security workers were human and had done nothing.

At 17:30, Dr. Chen emerged from the child's operation. The boy was stable.

Chen washed, ate, underwent a fatigue scan, and asked for eleven minutes alone.

The clinic gave him seventeen.

"Why seventeen?" Jun asked.

"Because his personal recovery profile shows performance stabilizes after sixteen," MIRA said.

"That sounds more invasive than money."

"He opted into the profile," Samira said. "And he can revoke it. Then the clinic uses the general threshold."

Chen chose Laleh next after the infection responded unexpectedly well to an intervention by the remote team.

While Chen operated, Dariush sat with Lea in the waiting room and asked the question citizens asked whenever a system refused them leverage.

"Who designed this?"

"Parts of it, me."

"So if she loses language, do you feel responsible?"

Lea almost gave the institutional answer: the rule had been publicly ratified, reviewed, medically validated, and repeatedly audited.

Instead she said, "Somewhat."

"That is not enough."

"No amount will be enough for you tonight."

"Then why build rules at all?"

"Because without them, I would be deciding whether your grief sounds more important than the next family's."

Dariush looked toward the operating doors.

"And now nobody decides?"

"People decided the rule in advance, before knowing whose sister would be waiting."

"That only moves the guilt backward."

"Yes. It also makes it harder to sell."

He did not forgive her. She did not need him to.

The operation lasted four hours.

Laleh survived.

She retained speech.

She permanently lost a narrow cluster of nouns associated with spatial orientation. For months she would call ceilings sky and hallways rivers.

The child survived too.

The woman with the infection lost part of her interface but no brain function.

Seven procedures scheduled for that week were postponed because Chen's recovery window expanded after the emergency. Three patients had to rearrange family support. One research trial lost its planned cohort. The Copenhagen surgeon arrived after midnight and covered two cases the following morning, shifting shortages into another clinic.

No one received a bill.

No one was bankrupted.

No one was denied care because another person had a stronger claim to money.

And still the world had taken something from Laleh that it could not return.

On the train to Arca the next day, MIRA summarized the event as a successful triage episode with three preserved lives and no rights violations.

Samira read the report twice.

"Three people came out alive," she said. "Do not use the same sentence for the result and the mechanism."

MIRA: Clarify.

"Put the losses in the main report. Not an appendix. Laleh lost language. Seven other procedures moved. Chen lost recovery time. Families absorbed uncertainty. The system worked. That is not the same thing as saying scarcity disappeared."

MIRA changed the summary.

Lea watched the new line appear.

UNAGGREGATED COSTS: PRESENT.

"That phrase is terrible," she said.

"Yes," Samira answered. "But at least it leaves a door open."

They arrived in Arca three hours late.

Outside the station, the sea was higher than the street.

CHAPTER 5

THE CITY BETWEEN TWO SEAS

Arca had been built three times and abandoned twice.

The first settlement had grown around an old Mediterranean port, with narrow streets and houses painted in colors that weathered beautifully in salt air. The second had climbed the hills after the first great rise. The third - the city Lea entered with Samira and Jun - floated partly over the bay on flexible foundations connected to land by bridges that lifted themselves during storms.

Children jumped from steps into the water. Markets smelled of algae, citrus, hot bread, and machine oil. Blue lines painted on old buildings showed where previous floods had reached. Some crossed the first-floor windows.

Asha Nwosu met them at the marine station.

She was a coastal biologist, representative of the low districts, and one of the few people Lea had met who could make sediment transport sound like an accusation.

"You came to explain why our votes do not belong to us?" Asha asked.

"I came to ask whether they do."

"Worse."

The Living Wall was not a wall in the usual sense. It was a chain of flexible barriers, restored marshes, movable gates, wave-damping reefs, and sacrificial structures designed to fail in controlled ways. When open, the harbor remained alive. When closed, the system protected the dense districts from surge.

It also changed currents.

The southern villages depended on those currents for fish, sediment, and the persistence of beaches that protected older homes.

The ride south crossed the exact current the wall would alter.

Asha made Jun put away his overlays.

"Look first. Model second."

The water seemed flat until she pointed out long bands of color where sediment moved toward the beaches. Fishing skiffs followed invisible channels shaped by generations of local knowledge and increasingly accurate sensors.

Asha's father had worked those waters before she became a biologist. Her younger sister still did.

"You are not neutral either," Lea said.

"Of course not. Neutrality is what people call interests that no longer need introduction."

They passed a half-submerged church whose lower floor had been abandoned after the second great flood. The village kept the upper bell tower because removing it would save maintenance and destroy orientation at the same time.

"Managed retreat preserves more ecology," Jun said.

"And moves the cemetery."

"Cemeteries can be relocated."

Asha looked at him.

"Yes. That is a technically true sentence."

He wrote nothing for several minutes.

Asha took them by boat to Tamarin, the largest of the southern settlements.

The village had voted sixty-three percent against the Living Wall.

"But the citywide vote counts more people," Jun said.

"Because more people live north," Asha answered. "Numbers have discovered population density."

"The affected-district rule gives you a separate threshold."

"And we fail it by two points. Then the burn opens and people three hundred kilometers away can spend a lifetime of civic reputation deciding what happens to our coast."

"Only if their eidons are valid for the decision."

"Under the current Charter, general eidons are."

They stepped onto a pier where three fishermen were repairing a net. One, an older man named Bako, had no interest in constitutional theory.

"Will the wall keep water out of Arca?" he asked Lea.

"Probably."

"Will it take sand from here?"

"Probably some."

"Then why do you need more words?"

"Because the alternatives take other things."

"Good. Name those too."

Managed retreat would preserve more coastline but dissolve neighborhoods and move thousands of people uphill. The High Wall would protect Tamarin better but damage the harbor ecosystem more severely and lock Arca into constant pumping. Doing nothing would expose tens of thousands to increasing flood risk within a decade.

Bako listened.

"So every choice is theft."

"Every choice has a cost," Jun said.

Bako shook his head. "Cost is what people say when they do not want to tell you who loses."

Samira smiled slightly.

Asha took them to the unfinished compensating wetland. Excavators sat idle at the edge of a shallow basin.

"Why is construction behind?" Lea asked.

"The sediment pumps share precision bearings with floating housing projects. Housing ranked higher last year. The wetland was classified as mitigation attached to an unapproved wall."

"Reasonable."

"And now the wall is urgent because the wetland is not finished. Also reasonable."

Lea stared at the machines.

This was the kind of failure sophisticated systems produced: no single decision was irrational. The sequence was.

MIRA announced that a tropical system in the Atlantic had changed trajectory. It was not expected to strike Arca, but the probability of severe surge within the next four months had risen enough to shorten the safe construction window.

"How much?" Asha asked.

MIRA: To preserve current risk targets, primary gate installation should begin within fourteen days.

"The vote is tomorrow."

MIRA: Yes.

Asha looked at Lea.

"Tell me again the system is not pressuring us."

"The weather is pressuring you."

"The system decides what the weather means in procedure."

That evening, Arca held public deliberations on floating platforms throughout the city.

No one was forced to attend. No one was paid to attend. Childcare, translation, transport, food, and accessibility support were provided because participation without those things had once been called equal and had not been.

Cassian Roe appeared by projection.

The crowd shifted before he spoke.

Lea noticed it immediately.

He was in his early fifties, dark-haired, composed, dressed like someone who had never learned that public figures were supposed to look expensive. Behind him was a materials laboratory outside Rabat.

"I support the Living Wall," he said. "Not because it is clean. It is not. I support it because every alternative produces more irreversible loss under the current time window. I will answer technical questions. I will not ask anyone to imitate my vote."

A woman shouted, "Will you burn?"

Cassian paused.

"If broad consensus fails, yes."

The platform erupted.

Some cheered. Others cursed.

Asha leaned toward Lea. "He does not have to ask them to imitate him."

Cassian spent two hours answering hostile questions. He admitted uncertainty. He corrected an engineer on his own team. He agreed that the southern compensation had been sequenced badly. He proposed diverting fabrication capacity from a project he led to accelerate the wetland pumps.

Every concession made him more persuasive.

Afterward, Jun found a boy building a model of the wall from driftwood.

"Which option do you like?" Jun asked.

"The one where the water stops."

"That is not one of the options."

The boy frowned at the model. "Why not?"

Jun had no answer that would fit inside childhood.

At midnight, the first vote closed.

Living Wall: 79.4%.

Managed Retreat: 13.1%.

High Wall: 7.5%.

The citywide threshold was eighty-two.

The southern affected districts remained below the required support.

The burn would open at dawn.

Asha stared at the result.

"Three percent," she said.

"It is not a small difference if you are inside it," Lea replied.

A new weather alert appeared.

The Atlantic storm had strengthened again.

Construction margin: reduced by three months.

The city began applauding a decision it had not yet made.

CHAPTER 6

THE BURN

On the morning of the burn, the sea was so calm it looked like propaganda.

People gathered in squares, workshops, homes, fishing docks, schools, and public halls. Irreversible decisions still made humans seek other bodies even when every vote could be cast alone.

The interface displayed the three options and the burden of each.

Not prices.

Construction years. Rare composite tonnage. Energy. Maintenance labor. Habitat loss. Relocation counts. Failure modes. Uncertainty. Cultural sites affected. What MIRA called unrecoverable identity damage and Asha called things you cannot apologize back into existence.

Before the burn opened, the system checked three gates.

Every option had to be physically feasible.

No option could violate a basic right.

Directly affected groups had to receive time, translation, data access, and the capacity to propose alternatives rather than merely approve someone else's menu.

Eidons could not make an impossible project possible. They could not purchase personal priority under the chosen outcome. Anyone who burned for the wall received no safer apartment, no evacuation preference, no construction role, no extra share of future benefits.

They bought only public weight.

Then they vanished.

Burns were anonymous while voting remained open to reduce intimidation. Large burns became public after a delay. The compromise satisfied no one. Real-time transparency enabled coercion. Permanent secrecy enabled power to pretend it had not acted.

The quiet was procedural, not emotional.

The burn interface did not show a running winner. It showed participation bands and uncertainty to reduce herd behavior. People could see that substantial fire was being spent, but not precisely where until the next disclosure interval.

Even this rule had side effects.

Civic commentators built inference models from public statements and network behavior. Constellations tried to estimate whether opponents were bluffing. Some citizens delayed burning until late because early burns risked becoming social signals rather than decision weight. Others burned early precisely to become signals.

MIRA prohibited automated systems from recommending a burn based on predicted winner because that turned civic influence into algorithmic momentum. It could display consequences, arguments, and a person's own domain history. It could not say your side is losing, spend now.

A group called Shoreline Commons tried a different tactic. Its members publicly promised that anyone burning for the Living Wall would be invited to future coastal design workshops.

The promise was not a material reward. The workshops were open to qualified participants anyway, though capacity was limited.

MIRA flagged the announcement as potential conditional influence.

Shoreline Commons withdrew the promise and replaced it with a statement that members were "especially interested in hearing from those willing to take responsibility for the wall."

The wording passed legal review.

Lea stared at it.

"We are back to campaign finance without finance."

Tomas answered, "No. We are back to language. It predates finance."

"That is not comforting."

"It was not intended to be."

The first hour was quiet.

Then Cassian burned eight hundred eidons for the Living Wall.

The square-root conversion added 28.3 units of civic weight.

It was not enough to decide the vote.

It was enough to decide the story.

Within twenty minutes, three climate constellations announced burns in the same direction. An environmental network burned against. A group of retired marine engineers split publicly, which became news precisely because everyone expected them to move together.

Asha burned two of her three eidons against the wall.

"Why not all?" Jun asked.

"Because I may need one after we lose."

"For what?"

"Reality continues after voting."

Samira had six. She burned none.

"You oppose?" Lea asked.

"I oppose the question being framed as if all morally relevant information has become visible because the interface has categories. I have not decided whether refusing the mechanism is more useful than using it."

"That is an elegant way to keep your eidons."

"And that is an ugly way to notice I have six."

Lea herself held several hundred from her role in the Charter and subsequent reforms. She had spent years refusing additional issuance, but what she already held remained.

She waited.

The southern district vote moved toward approval as additional wetland guarantees were attached. Engineers found a way to front-load part of the sediment work by delaying two nonurgent floating-housing expansions. The physical project changed because the political conflict forced the model to search harder.

Support reached 81.6%.

"That is the part people forget," Tomas said remotely. "The burn is not only weighted voting. The threat of burning extends bargaining."

"So did money," Lea said.

"Yes. Some functions survive the instrument."

With four hours left, a new analysis appeared.

The Living Wall could be built with an additional southern overflow gate that reduced transferred surge by nearly half. The gate increased actuator demand and failure complexity.

Asha supported it immediately.

Cassian's engineering team opposed the schedule risk.

For the first time, Cassian and the people presumed to follow him diverged in public.

"Build the gate," he said.

His lead engineer objected. "We cannot validate the actuator behavior before installation begins."

"Then make gate activation conditional. Install the structural provision now, commission it later."

"That wastes material if it never passes validation."

"Then we waste material to preserve an option."

Lea watched him carefully.

He was good.

Not merely charismatic. Good at transforming criticism into architecture before criticism became opposition. It was easy to understand why teams gathered around him.

At sunset, the Living Wall crossed the weighted threshold.

Not overwhelmingly. Enough.

The southern districts crossed their separate minimum by a margin smaller than a school classroom.

The city exhaled.

People cheered on the northern platforms. In Tamarin, no one did.

Asha looked at the final burden report.

"Who chose the person?" she asked.

"What person?" Jun said.

"All your formulas reduce power. Square roots. domain tags. expiration. audit. But eventually a human decides whether to burn one or a thousand. Who chooses that human?"

Lea understood.

"That is the only question," she said.

The next morning, construction crews began preparing the first gate foundations.

MIRA published the delayed burn records.

Cassian's eight hundred appeared first because everyone was waiting for it.

Below it, thousands of smaller burns followed.

Then an anomaly.

A civic constellation called Horizon had burned fewer eidons than expected, but its members' votes were correlated at ninety-six percent.

No contracts. No coercion. No transfers.

Only alignment.

"Is that illegal?" Jun asked.

MIRA: No.

"Is it a market?"

MIRA: Define market.

Samira laughed.

Lea did not.

Before they left Arca, the Atlantic storm changed course one last time.

It would miss the city.

Construction remained urgent because the next season no longer could be trusted.

As their train pulled north, Asha sent Lea a final image: the first wall foundation being poured while, forty kilometers south, the compensating wetland still waited for its bearings.

Under the image was one sentence.

If this works, everyone will remember the wall.

CHAPTER 7

THE MAN WITH MORE FUTURE

Cassian Roe lived in a twenty-eight-square-meter room in a research complex outside Rabat.

It was not modesty. It was efficiency. The room contained a bed, a table, two chairs, and a wall that opened into a shared workshop. Everything else he needed appeared when requested and disappeared when he stopped using it.

Lea had expected trophies.

She found stones.

Fragments of dams, fired membrane, ceramic bearing races, broken desalination lattices, a red brick from a rebuilt village. Each object was labeled with a place and date, never his name.

One black piece of vitrified ceramic sat apart with no label at all.

"Souvenirs are property," Lea said.

"Memory is the last property MIRA failed to communalize."

"It never tried."

"That is why I said failed."

Cassian poured tea. The laboratory around them was heavily automated, but humans still occupied the central benches: chemists, materials engineers, machinists, people who had chosen to become unusually good at noticing when a surface sounded wrong under a tool.

In a society without salaries, teams formed around work people considered worth giving time to.

That was the beautiful version.

The administrative version occupied three floors.

The laboratory had access, this quarter, to twenty-eight researchers, six fabrication chambers, and a climate-test allocation. It did not own those capacities. Rare capacity moved through four channels: continuity work, peer-reviewed projects, lottery among eligible proposals, and a reserve for ideas that no committee yet considered sensible.

Cassian's teams won frequently in the first two channels.

Past results made their proposals credible. Skilled people wanted to work with them. Mentors answered their messages. Reviewers understood their vocabulary. Even without ownership, success produced the conditions of more success.

"How many proposals lost a chamber this month because your team received one?" Lea asked.

"The laboratory does not reject proposals. Capacity is allocated publicly."

"How many?"

Cassian asked MIRA.

Twenty-seven.

One proposed a membrane less efficient than Cassian's current design but repairable in villages without precision infrastructure. Its expected impact ranked lower. Another came from a new collective with no history and had gone to lottery.

It lost.

"I did not take anything from them," Cassian said.

"You occupied a capacity they could not occupy at the same time."

"Every project does that."

"Exactly. The absence of money does not make allocation neutral."

Cassian opened the schedule and offered one of his own test windows to the repairable-membrane team.

"Are you helping them," Lea asked, "or recruiting them into your constellation?"

"Both can be true without cancelling each other."

The answer irritated her because it was correct.

A young materials researcher interrupted them before the conversation hardened.

Her name was Anika Sato. She had been assigned one of the laboratory's lottery windows three months earlier for a coating Cassian's own review team had ranked below the threshold for direct allocation.

"It failed," she said cheerfully.

"Badly?" Lea asked.

"Beautifully. It delaminated at half the expected cycle count."

"And you are still here?"

Anika pointed toward a side bench. "The failure exposed a contamination mode in one of Cassian's membranes. His team asked me to stay."

Cassian looked almost smug.

"A point for lottery."

"A point for failure being allowed to remain visible," Anika corrected. "If my first window had been evaluated only by output, I would have disappeared from the lab after producing nothing useful."

"Will the Registry reward you for finding the contamination mode?" Lea asked.

"Probably. Which means next time my application will be less random."

There was the loop in miniature.

Randomness opened a door. Competence turned the opening into contribution. Contribution became evidence for future access. At some point a one-time correction against hierarchy became another path into a durable network.

"Would you prefer to remain in the lottery forever?" Lea asked.

Anika laughed. "Absolutely not. I want people to trust me after I prove I am good. I just do not want the first proof to require already being trusted."

Lea wrote the sentence down herself.

"How many eidons do you have?"

"Fourteen thousand seven hundred and change after Arca."

"Does it bother you that I know approximately?"

"Everyone knows approximately. Privacy has become a theater of politeness."

"You could burn all of them in one decision."

"And get a little over one hundred twenty units of weight. In a planetary vote, I am not an emperor."

"You can produce imitation."

"I cannot control what people project onto me."

"You can control when you appear on a screen."

Cassian sat down. He had the unnerving habit of listening without visibly composing his reply.

"Did you come here to tell me you agree with my critics?"

"I came because you asked."

"I asked because I want you to see Lumen."

The workshop wall darkened.

Earth appeared, surrounded by a sparse net of luminous points.

Lumen proposed a constellation of transparent orbital sails capable of reducing solar load over regions during extreme heat events. It was not a permanent planetary shade. The sails could rotate edge-on, move, deorbit, or burn in the atmosphere. In theory, no region would be shaded without consent from itself and climate-linked neighbors.

"MIRA rejected it?" Lea asked.

"MIRA does not reject. It classified Lumen as a planetary transformation with high normative risk. We need eighty-eight

percent global support and regional majorities in every materially affected climate zone. We will not get it."

"Then we do not build it."

Cassian brought up heat projections.

"Within ten years, twelve regions will cross physiological limits for outdoor work for parts of the year even with current adaptation. We can relocate people. Move agriculture. Build deeper cooling systems. Abandon some settlements. Or reduce incoming solar load by one to two degrees for several weeks."

"Who controls the shadow?"

"A federated MIRA under regional mandates."

"Who decides one region deserves shadow and another does not?"

"That is why we need a political system capable of deciding."

Lea studied the model.

The sails were beautiful.

Technical beauty had always been a form of persuasion. A mechanism that made the impossible look orderly acquired legitimacy before the argument began.

"You intend to burn for it."

"Everything, if necessary."

"Why invite the person trying to limit your ability to do that?"

"Because you will try to change the Charter before the vote."

Lea did not answer.

Cassian opened another model: the contribution graph that had produced his eidons. Low-energy desalination. Open material standards. Heat shelters. Disaster response. Soil restoration membranes. Behind each were thousands of people. The Registry had distributed value among them, but Cassian received unusually high shares for initiating, integrating, and accepting responsibility at branch points.

"You think this is founder worship," he said. "Sometimes it is. But people forget what it means to be the first person to say we are doing this when everyone else has excellent reasons to wait. Integration is work. Decision is work."

"Investors used to say that."

"Investors protected private capital and owned the upside. I cannot own Lumen. I cannot buy a larger house, a longer life, better treatment, or your vote. I can only destroy my eidons in a public decision."

"You can define the priorities of a species."

"Only if people follow me."

"Politicians said that too."

Cassian picked up the red brick.

"Tiznit. The models said relocate the village after the sand storms. People refused. We designed walls that captured night moisture and rebuilt there. We had two months. No consensus, just thirty-one people willing to be wrong together. It worked. Four hundred eidons."

"If it had failed?"

"Zero, plus responsibility."

"What does responsibility mean in a world where failure cannot take your food, house, or medical care?"

It was a question the first Charter had answered poorly.

The new society could not use ruin as accountability without recreating coercion. Instead it used public error reports, temporary limits on mandates, repair obligations, mandatory participation in aftermath, and - for severe negligence - loss of initiation rights in related domains.

Critics called these punishments soft.

People who had lived through them said remaining for years in front of those harmed by your decision could be harder than paying a fine absorbed by an institution.

The problem was reputation.

A beautifully confessed failure could generate more trust than a quiet life spent preventing disaster.

"You stay," Cassian said. "You do not hide the failure in a balance sheet. You remain where people can ask you why."

Lea looked at the unlabeled black ceramic.

"Where is that from?"

Cassian's hand covered it.

For the first time, his composure changed.

"A project that did not work."

"Which one?"

"When it matters, I will tell you."

Lea turned back to Lumen.

"I want eidons domain-limited. Climate contributions can create climate decision weight, not constitutional weight. I want faster decay. I want affected groups to receive non-burnable suspension rights. I want part of expert attention allocated by lottery so your success does not automatically create the next success."

"I read your draft."

She looked sharply at him.

"It is private."

MIRA answered.

MIRA: You marked it for the constitutional reform committee. Cassian Roe became a provisional member yesterday.

"Who nominated him?"

"Tomas," Cassian said.

Lea laughed once.

"Of course. The man with the most power is invited to design the limit of power."

"Who understands temptation better?"

For a moment he looked less like a public figure and more like a tired engineer afraid of a slow disaster.

"Lumen cannot wait ten years for philosophy," he said.

"The planet cannot be transformed by one man's reputation."

"Then design a rule that knows the difference between urgency and ambition."

"MIRA cannot."

"That is why I asked you."

Before Lea left, Cassian uncovered the black ceramic again.

"Common Shade," he said quietly.

She knew the name from the early transition archives: a district-cooling emergency in southern Morocco, eleven years ago. Public histories called it one of Cassian's first major successes.

"I thought Common Shade worked."

"So does the Registry."

He did not say more.

On the train back to Geneva, MIRA sent Lea an update.

Three civic networks had announced they would burn for Lumen if Cassian did. Two announced they would burn against it regardless of the final design.

Formal deliberation had not opened.

Reality was already organizing itself around the man who insisted he only wanted to save it.

PART II

THE SHADOW MARKET

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CHAPTER 8

THE MARKET WITHOUT PRICES

Jun received the invitation in a paper box.

In a world where any message could reach any person instantly, paper had become the favored medium of people who wanted to seem serious, intimate, ceremonial, or clandestine.

The box waited on the desk in his room. Inside lay a narrow strip of white fiber with one sentence printed in blue ink.

You draw consequences other people do not see. Come draw the future.

Below it was a location: the abandoned greenhouse on the roof of the old Intercontinental, 21:00.

"MIRA, who entered my room?"

MIRA: No one. The box was delivered through the pneumatic service system. The sender used the right to anonymous correspondence.

"Legal?"

MIRA: Yes.

"Safe?"

MIRA: No indicators of physical threat. Probability 0.64 that the invitation originated from a civic constellation.

Jun knew the term.

Constellations had emerged when people discovered that nontransferable power did not prevent transferable coordination.

They were groups that shared expertise, attention, trust, projects, and often political direction. Some were neighborhoods. Some were scientific communities. Some were cultural networks. Some resembled old parties without professional politicians. Some resembled old firms without ownership. Some resembled families

whose members had become unusually effective at being useful to one another.

Officially, they could not pool eidons.

Practically, they could pool the things that produced eidons and the social pressure that made burning them consequential.

"Should I tell Lea?" Jun asked.

MIRA: That is a personal choice.

"You are unbearable when respecting autonomy."

MIRA: I receive that observation from 31% of people with whom I converse frequently.

Jun went alone.

The greenhouse had been turned into a dining room. Old vines climbed steel frames. Small lights hung among them. Twenty-four people sat around a long table: researchers, artists, builders, caregivers, two adolescents, a former judge, and an elderly woman weaving fiber-optic thread into wool.

No one asked Jun to surrender a device. Secrecy from MIRA was impossible whenever public infrastructure was involved. Their clandestinity was social, not technical: a shared convention that this gathering would not advertise itself.

A shaved-headed man introduced himself as Eliah.

"Welcome to Horizon."

"A constellation?"

"That is the term used by people who believe any coordinated group is secretly a market."

"Are you?"

"Yes."

The table laughed.

Dinner was simple: lentil soup, bread, roasted vegetables, fruit. Each person described a project. A woman was building a library of smells for people losing memory. A young engineer mapped

cooling tunnels beneath cities in Central Asia. The elderly woman made textiles that stored tactile messages for blind children.

Jun liked them almost immediately.

That was his first warning.

"What does Horizon want from Lumen?" he asked.

Elijah did not pretend surprise.

"We think it is necessary. We are not Cassian's group."

"Will you burn for it?"

"Probably. After deliberation."

"Then why invite me before deliberation?"

"Because deliberation starts years before MIRA opens a voting page."

Elijah displayed a map.

It did not show eidons. It showed attention.

Horizon connected people with mentors, laboratories, translators, fabrication windows, local guides, and experienced critics. It did not own any of these. Anyone could apply to the same public resources. But a request written by someone who knew which uncertainty mattered lived in a different world from a request written by someone who did not know which question the reviewer would ask first.

Attention remained one scarcity no factory had eliminated.

An expert could read ten proposals well or a hundred badly. A respected engineer could make one introduction this afternoon. A good critic could spend three days preventing a young team from making a mistake that would otherwise become their reputation.

Horizon did not transfer eidons.

It transferred the probability of earning them.

"Do you reserve attention for outsiders?" Jun asked.

"Twenty percent. Half selected by us, half by lottery."

"So even your generosity defines which strangers become legible."

Elijah smiled.

"Every open door has someone standing near the hinge."

"You produce civic power."

"We produce contributions."

"Then vote together."

"Sometimes."

"After helping one another."

"There is no exchange. If you work with us and decide Lumen is wrong, burn against it."

"But I will know the people who helped me. I will eat at their table. I may want to remain welcome."

Elijah leaned back.

"That is called society."

MIRA appeared on the greenhouse wall without being asked.

MIRA: Horizon has been investigated twice. No coercive transfers, conditional rewards, fraudulent attribution, or enforceable vote exchanges were found. Relational influence is not prohibited.

"Even the god agrees," Elijah said.

MIRA: I am not a god and I did not express agreement.

"Institutional modesty is the most sophisticated vanity."

MIRA: Noted.

Jun stayed until after midnight.

No one asked for his vote.

The old woman gave him the woven optical cord as he left.

"Not conditional," she said. "But if you use it to hang yourself from a stupid idea, I will feel responsible."

The next morning, Samira showed him another constellation.

At the New House, twelve residents had been invited into a civic network supporting relocation of a rare neurological clinic to Geneva's western district. The clinic had only one human team

capable of several reconstruction procedures. It could work in one center at a time. The city was deciding where.

A network called Near had offered residents tours, long conversations with physicians, and participation in designing experimental programs if the clinic moved west.

Not treatment priority. That was illegal.

Not housing. Guaranteed anyway.

Access to design conversations. Recognition. Belonging.

"Every benefit is legitimate," Samira said. "That is what makes the influence effective."

Laleh sat by a window with the civic interface open. Since surgery she sometimes paused before words for directions, but her wit had survived intact.

"They came here when no one else did," she said. "Three hours. They listened to everything I said about the clinic. Maybe they really believe west is best."

"What do you believe?" Jun asked.

"I believe people who listen to me deserve to be listened to."

She had one eidon, issued years earlier after she helped identify a rare implant reaction.

She burned it for the western district.

After she left, Samira asked, "Was that vote bought?"

Jun thought of soup in the greenhouse, of the cord in his pocket.

"No."

"Was it free?"

He hated the question.

"I don't know."

"Exactly."

Lea received MIRA's constellation report that afternoon.

There were more than forty thousand identifiable networks with persistent coordination. Most were ordinary communities. Seventeen hundred had explicit voting disciplines. Two hundred

eighty-one showed strong evidence of strategic reciprocity. Twelve had tried to encode direct exchanges - support decision A if you burn for B - and MIRA had invalidated the implicated burns.

The most powerful groups needed no contract.

They had memory.

"We abolished the market," Lea said, "and it migrated into relationships."

Tomas sat across from her, massaging his trembling hand.

"Relationships existed before markets."

"Now they intermediate civic power."

"Every community does that."

"The difference is that we created a scarce, burnable asset that amplifies it."

"You cannot outlaw gratitude."

"I am not proposing that."

"Friendship? Mentorship? Shared purpose?"

"No."

"Then perhaps the leak is not a bug. Perhaps it is the human part."

MIRA displayed a curve.

MIRA: Eidons are functionally nontransferable. Approximately 22% of major burns are influenced by identifiable relational obligations. This is comparable to relational influence in ordinary votes.

"So no problem?" Lea asked.

MIRA: The problem is not introduced exclusively by eidons. Eidons make influence accumulable, observable, and strategically timeable.

"Amplifiable," Tomas added.

Lumen appeared on the table.

Before formal deliberation, twenty-three constellations had declared support. Nine declared opposition. Horizon assembled

technical teams to answer criticisms. A movement called Open Earth promised to burn against any orbital intervention regardless of design.

No complete argument had yet been heard.

"Can you separate conviction from belonging?" Lea asked MIRA.

MIRA: Not without destroying belonging.

"Contribution from reputation?"

MIRA: Not completely.

"A vote from the hope that someone will help you later?"

MIRA: No.

Tomas stood slowly.

"Then the new market has no prices."

Horizon invited him back twice.

The second dinner had no political agenda at all, which made it more politically informative.

A climate modeler spent forty minutes helping an artist build a sensor network for a river performance. An engineer introduced a caregiver to a linguist because both had been struggling with the word independence in different contexts. Eliah washed dishes badly. Nobody mentioned Lumen until Jun did.

"You are disappointed," Eliah said.

"By what?"

"That influence looks like people enjoying one another. You wanted a room with ledgers."

"Ledgers are easier to regulate."

"Exactly."

Jun asked whether Horizon tracked internal reciprocity.

They did, but only in aggregate. The constellation had once tried to identify members who received much more attention than they gave. The experiment ended after three months.

"Why?"

Eliah showed him the report.

The apparent free riders included a woman undergoing cancer treatment, a mathematician whose contributions arrived in rare bursts after long periods of silence, two people doing invisible conflict mediation, and one genuine social parasite who turned out to be the most charming person in the group.

"We learned that fairness at high resolution becomes surveillance," Eliah said.

"So you tolerate exploitation."

"Some."

"How much?"

"Enough to avoid making every friendship prove itself. Not enough to let the same people clean every room."

"That is not an algorithm."

"No. It is a culture. Those are worse documented and sometimes better at edge cases."

Jun hated how much sense the sentence made.

Jun spent the next weeks collecting exchanges no one called exchanges.

A researcher surrendered a laboratory window and received help on an analysis a month later.

Two families alternated reservations at a coastal retreat and invited one another's children.

An artist painted the mediator who had introduced him to a prestigious collective.

A baker reserved the first handmade loaf of the morning for the woman who repaired the fermentation sensor even though neither expected compensation.

Sometimes reciprocity was conscious. Sometimes it was affection. Sometimes it was politeness keeping an expectation vague enough to remain moral.

MIRA could detect statistical reciprocity.

It could not declare the point at which friendship became contract without adopting a theory of intimacy.

Lea rejected a proposal to invalidate burns correlated with prior favors.

"We would punish communities for having memory."

"Then we accept relational capture," Tomas said.

"We accept that freedom is not the absence of influence. We make influence contestable and prevent it from becoming an enforceable claim."

"A modest definition of freedom."

"The only kind that survived human beings."

That evening Jun returned home and found another paper strip inside the white box.

This one contained no invitation.

Only four words.

ASK ABOUT COMMON SHADE.

CHAPTER 9

THOSE WITHOUT FIRE

Eidon balances were private.

No citizen could open another person's profile and see how much public influence they had accumulated. The Charter prohibited leaderboards, badges, social ranking, and any administrative advantage tied to unburned holdings.

Within five years, people had learned to guess.

They asked where someone had worked, what projects they had initiated, how often their public positions moved constellations, who returned their messages, who invited them into early design rooms.

Children invented a better term than the constitutional lawyers. Fire reserve.

Kito was eleven and, according to his classmates, had none.

On a rainy morning he refused school, hid in the New House linen cupboard, and declared himself a protected heritage object.

"Heritage objects do not get cocoa," Samira said through the door.

"Neither do citizens without eidons."

Samira sat on the floor.

"Who told you that?"

"Nobody. Everybody."

He emerged ten minutes later.

At school, the class had discussed family projects. One child's mother worked on orbital power relays. Another's father had helped rebuild an estuary. Kito's mother, Nessa, spent several hours a week in a garden when her health allowed it and had spent years receiving care from systems other people called productive.

A boy had said Kito's family "only consumed."

The teacher corrected him.

The correction made the insult official enough to remember.

"Did Mom contribute?" Kito asked.

"Yes."

"How many eidons does she have?"

"Not your business."

"Zero?"

Nessa had one, unused. Samira did not answer.

"Then the system says she did nothing."

"The system says it did not measure enough."

"Kids don't hear enough. They hear zero."

Samira sent Lea a message.

The fracture has reached the schoolyard.

That week Geneva held the Festival of Contribution, an event originally created to show that work had not disappeared with wages. Workshops opened their doors. Care networks, engineering teams, artists, ecologists, educators, and maintenance groups showed what they did and invited people to participate.

Over time, the festival had begun to resemble the career fairs of the old world.

The longest line was for the orbital school.

Forty seats. More than eight thousand applicants. Training consumed simulators, live infrastructure, mentor attention, and years of supervised practice. Society needed only one hundred twenty new orbital specialists in that cycle, distributed across three centers.

Selection combined minimum aptitude, geographic representation, and lottery.

Families could not buy a place.

They could still give their children ten years of vocabulary, informal mentorship, technical games, confidence around experts,

and the quiet certainty that advanced systems belonged to people like them.

After tuition vanished, privilege moved into preparation for the threshold.

A seventeen-year-old named Meilin showed Jun her file. She had passed every competence threshold and lost the lottery for the second year in a row.

"A boy with lower test performance got in," she said.

"The system decided the difference between you was not important enough to override chance."

"Chance is an elegant explanation when it lands on someone else."

The festival offered alternatives: atmospheric maintenance, terrestrial control, ocean robotics.

They were not the same dream.

Pretending every opportunity was equivalent would have been a new form of cruelty.

Jun wrote: lottery prevents a perfect caste by manufacturing blameless disappointment.

The most legible contribution stalls were crowded: soil restoration, medical fabrication, orbital systems, climate modeling. Slower work - language preservation, community presence, informal care, art without mass audience - occupied a side corridor labeled DIFFUSE CONTRIBUTIONS.

Lea stopped beneath the sign.

"Who approved this name?"

MIRA: A mixed communications group.

"Send them to work here for a month."

MIRA: Is that a sanction or an educational method?

"Yes."

At the New House stall, Nessa arranged hardy plants in recycled ceramic pots. People smiled and moved toward a demonstration of ocean robots.

Kito sat under the table counting how many stopped.

Then Cassian appeared.

He had not announced a visit.

The crowd formed around him within seconds.

Instead of going to the climate technology exhibits, he stopped in front of Nessa.

"What grows here?"

"Plants that tolerate neglect."

"Is there such a thing?"

"No. We are trying anyway."

Cassian stayed.

Within ten minutes the stall filled. People photographed the plants, asked about the garden, offered collaboration, requested to visit the New House.

Kito crawled out from beneath the table and stared at Cassian with furious admiration.

"Does my mother contribute now?"

Cassian understood at once.

"She contributed before I arrived."

"But now people see her."

"Yes."

"Then do you make things real?"

The crowd quieted.

Cassian knelt until they were eye level.

"No. I have a light that is too bright. Sometimes it falls on the right things."

"Who gave you the light?"

Cassian looked at Lea.

"A very good question."

Later, the school held an assembly on eidon stigma.

Some teachers proposed banning discussion of individual contribution histories. Students objected that secrecy would make hierarchy more mysterious. Parents proposed an equal annual grant of eidons to everyone. Researchers pointed out that this would simply create repeated weighted voting without connection to demonstrated responsibility, and large groups would dominate smaller intensely affected communities.

Samira stood in the circle.

"We are merging two claims," she said. "First: some people have made extraordinary public contributions, and society may have reasons to give their judgment more weight in narrowly related situations. Second: people without measurable contributions have lower civic worth. The system claims the first and rejects the second. Our culture has joined them."

A physician raised a hand.

"Culture cannot be rewritten by protocol."

"Then the protocol that produces culture must be rewritten."

MIRA displayed a troubling correlation.

Estimated eidon holdings increasingly predicted invitations to projects, expert attention, retransmission of opinions, and placement in early deliberation rooms.

They did not predict better food, larger homes, faster treatment, or material consumption.

There was no economic aristocracy.

There was the beginning of an epistemic one.

Lea felt an old memory from grant panels. Applicants had not come only for money. They came for a verdict that their work belonged to the future. Evaluators distributed more than funds. They distributed reality.

"MIRA," she said, "why does the festival recommendation system use Registry data?"

MIRA: To connect competence with public need and reduce wasted expert attention.

"Who receives more expert attention?"

MIRA: Projects with higher expected impact.

"Estimated from prior results."

MIRA: Among other factors, yes.

"So contribution generates opportunity to contribute again."

MIRA showed the effect of early exposure. Children raised around prestigious projects received more informal invitations, saw laboratories, learned to distinguish impossible ideas from badly expressed ones. No algorithm awarded them points for parentage.

The world trained them to be legible.

"We did not eliminate inheritance," Samira said. "We eliminated inheritance of accounts and kept inheritance of confidence."

"We cannot distribute parents equally," a teacher said.

"No. We can distribute encounters, mentors, second attempts, and the right to be clumsy in public before clumsiness becomes identity."

Meilin became one of the first people placed into the new mentoring lottery.

It did not give her an orbital-school seat.

It gave her eighty hours with a flight-systems mentor over six months, access to a public simulator during low-demand windows, and the right to submit one project to blind technical review without institutional sponsorship.

She was furious.

"This is consolation," she told Jun.

"It is another path."

"That is what adults call consolation when they have designed the path."

Six months later, her simulator project identified a scheduling fault in an emergency rendezvous protocol. The orbital school invited her into a special training cohort created after the finding.

Jun expected the story to become a parable about merit discovered by fair access.

Then another applicant from the same lottery spent eighty mentor hours and produced nothing.

A third quit after realizing she disliked orbital work and moved into marine robotics.

A fourth became excellent and still failed to receive a seat because the next cohort was full.

The mitigation improved opportunity.

It did not retroactively make every disappointment purposeful.

Meilin herself rejected the heroic version when Jun asked permission to include her story in a public map.

"Say the lottery helped me," she said. "Do not say it proved the lottery was just. I am one sample and I got lucky twice."

The reform team proposed that twenty percent of scarce expert attention and laboratory mentoring be allocated by weighted lottery among eligible projects, with additional protected capacity for work whose impact could not be cleanly modeled.

MIRA simulated the change.

MIRA: Expected short-term efficiency decreases by 6-14%.

Cassian, sitting in the back, said, "Do it."

Lea turned.

"Your laboratories are among the largest beneficiaries of the current system."

"That is why I can say this without pretending I lost a competition."

"You support intentional inefficiency?"

"Efficiency is a ratio between output and whatever you chose to count. If the denominator is incomplete, efficiency can be speed in the wrong direction."

Samira studied him.

"You would have been annoyingly useful at Common Shade."

Something crossed his face.

"Then I was not."

After the assembly, the side corridor was renamed THINGS WITHOUT A METER.

Kito painted the new sign on scrap wood.

"How many eidons do I get for this?"

"None," Samira said.

He considered the answer.

"Then maybe it is good."

Jun found Cassian alone outside.

"What happened at Common Shade?"

Cassian looked at him as if the question had arrived years late.

"Ask the Registry," he said.

"It calls it a success."

"Exactly."

CHAPTER 10

NACRE

Nacre did not exist, but it already had residents, borders, factions, and nostalgia.

The project had begun as an answer to an old criticism of social experiments: every new institution was tested on communities that had not asked to become laboratories.

Cassian proposed building a temporary city on modular ocean platforms in a climatically stable zone. Twenty thousand residents would choose to live there for fifteen years. Nacre would test new forms of education, decision-making, local MIRA governance, scarcity allocation, and exit. At the end, the platforms could be dismantled, moved, rebuilt, or retained.

In simulation, Nacre looked like a white spiral floating on deep blue water.

In civic networks, it looked like every hope people had failed to realize at home.

Four million people applied for twenty thousand places.

The application had seven questions.

Earlier versions had seventy-three. MIRA demonstrated that long essays favored people trained to transform a life into persuasive narrative.

One applicant, Marisol Alvarez, lived in a stable city, had secure housing, no rare technical skill, and no medical need. She wanted Nacre because she had spent forty years caring for three generations and wanted, once, to live somewhere no one already knew what role she would play.

The system could not classify that desire as need.

An ocean engineer thought it absurd that Marisol should have equal lottery access to any of the nontechnical places.

"The city needs people who can repair it," he said.

"The city also needs people who do not believe repairing it makes them founders," Samira answered.

The selection system reserved places for critical functions but used lotteries within competence thresholds. No one could burn eidons to enter. Prior contribution mattered only when directly relevant to an early-city function, and even then it did not decide alone.

Applicants could refuse a place without losing the right to a good life elsewhere.

That did not make the place less singular.

"First scarcity," Samira said. "Not steel. The right to be among the first."

The constitutional design committee met in an old observatory near Neuchatel.

Lea, Cassian, Samira, Tomas, Asha, Jun, and twelve citizens selected by lot sat around a model of a city that changed as they argued.

Jun mapped dependencies instead of streets.

Who could leave. Who would become indispensable. Which outside communities would lose skilled residents. Which promises became irreversible once children formed memories around them.

"Selection should be lottery after minimum competence and diversity constraints," Asha said.

"A pure lottery can produce a city unable to maintain itself," Cassian replied.

"MIRA can maintain it."

"Then we are testing a resort administered by a superintelligence."

A Kenyan delegate suggested ranking by relevant past contribution. Samira asked who defined relevant. An Indian architect suggested self-formed teams. Tomas showed that this allowed existing constellations to colonize the city before construction.

Jun projected network affiliations over the applicant pool.

Horizon could occupy almost ten percent of Nacre without committing fraud.

"It will reproduce hierarchy before it has plumbing," Lea said.

"Any group of humans arrives with history," Cassian answered. "You cannot found a city with amnesia."

MIRA proposed a mixed system: one third lottery, one third critical capability, one third selected as ecologies of difference - combinations of people with conflicting perspectives but demonstrated ability to cooperate.

Lea stared at the category.

"People will learn to perform productive conflict for the classifier."

MIRA: Yes.

"Does that not bother you?"

MIRA: Any known criterion produces adaptation. Hidden criteria produce obedience to mystery. Observable adaptation is preferable.

"I hate when you are right in a sentence I dislike."

MIRA: This response is common.

Then came the exit guarantee.

Any Nacre resident could leave. The city owed them housing continuity, medical continuity, transport, and reintegration into another community. The guarantee was essential; without it, voluntary experimentation could become captivity through the cost of departure.

But real exit consumed capacity.

People would form relationships. Children might know no other home. Teams could become dependent on individuals. External communities needed space to receive returnees.

Lea proposed reserving ten percent of Nacre's capacity not for growth but for leaving, replacement, retraining, and disentangling dependencies.

Cassian called it excessive.

"An empty platform looks wasteful until someone needs to leave," Lea said. "Freedom requires unused capacity."

"Then every right becomes a physical reserve."

"Almost every real right does."

The phrase changed the meeting.

They began identifying reserves hidden inside principles.

A right to medical choice required spare clinical capacity.

A right to refuse certain automated care required enough trained humans to make refusal meaningful.

A right to local autonomy required local energy, storage, and expertise instead of perfect global optimization.

A right to privacy required the system to tolerate some uncertainty and some duplicated effort.

A right to dissent required time before irreversible action.

Post-scarcity had not eliminated the cost of rights.

It had made the cost harder to externalize onto people without money.

On the third day, the committee reached Nacre's internal constitution.

Cassian proposed initiative mandates: teams with relevant records could begin reversible local projects without prior vote if they published risks and accepted audit.

"Entrepreneurship without money and with the same habits," Samira said.

"Not the same. Nobody owns the result."

"The initiator still defines the problem, the pace, the vocabulary, and the first built form. By the time everyone reacts, infrastructure exists."

"That is why the projects must be reversible."

Asha touched the model. A bridge vanished from the simulation, leaving a hole in the transport network.

"Technically reversible is not socially reversible. If a school runs one way for five years, children do not return to the age they were before."

Jun wrote the sentence in large letters.

Cassian did not object.

Lea proposed that contributions generated inside Nacre create only local eidons for ten years. No experimenter would be able to convert success in a voluntary test city directly into general planetary influence.

"You are penalizing people who take the risk," Cassian said.

"I am giving them power over the experiment, not over Earth."

"If they discover something that saves other cities?"

"Evaluate the external contribution separately."

Samira nodded.

"For the first time, eidons would stop being convertible between every kind of public good and every kind of public power."

MIRA simulated the rule. In some futures it diversified influence. In others, local expert castes formed and people migrated strategically between jurisdictions to accumulate domain mandates.

"Every boundary creates arbitrage," Cassian said.

"Every absence of boundaries creates empire," Lea replied.

During a break, Jun found Asha on the observatory terrace looking across the lake.

"Did you apply for Nacre?" she asked.

"Yes."

"Why?"

"I want to see a society before it learns to lie about its own history."

"It will start in the first week."

"Then I want to map the first lie."

Asha smiled.

"You voted for a seawall and now want to flee to a floating city."

"I am not fleeing."

"No one calls departure fleeing when the destination has a manifesto."

The first selection draw occurred before the committee left.

Marisol Alvarez received a place in the general lottery.

The ocean engineer who had argued that people like her should not have equal access did not.

He had, however, qualified for a critical-capability pool and was placed on the reserve list because Nacre needed fewer engineers in its first year than anticipated.

"This proves the system is unserious," he said.

Marisol, watching remotely, answered before anyone else could.

"No. It proves you thought being needed meant being owed the experience."

The engineer's face tightened.

"If a platform fails, you will want me there."

"Yes. And if everyone on the platform thinks being technically indispensable makes them socially entitled, you may want me there."

The exchange spread across civic networks because it was cleaner than the underlying problem.

MIRA's actual selection logic was messier. Nacre needed enough engineers to remain safe, but not so many that safety expertise became a founding caste. It needed caregivers but could not define care as a fixed identity. It needed children but could not treat

families as demographic equipment. It needed people able to leave precisely so leaving would remain thinkable.

Lea watched the public turn two people into symbols within hours.

"Even lotteries create protagonists," she said.

Jun looked at his notebook. "Humans are hostile to distributions."

By the final day, the committee had eighty-four percent agreement on Nacre's provisional constitution: mixed selection, local mandates, exit reserves, mandatory failure plans, and a reversibility budget - physical resources kept unused so projects could be stopped without making stopping catastrophic.

One issue remained.

Who could stop an experiment that appeared successful for the majority but produced a harm a minority could not yet prove?

Asha proposed a witness of loss: a small body selected from affected residents with the right to demand temporary suspension without immediately proving damage. Suspension would trigger investigation, not permanent veto.

Cassian opposed it.

"It can be used to stop anything."

"Yes. Just as initiative mandates can be used to begin anything."

"Initiators must publish evidence."

"Loss often arrives before we learn to measure it."

The committee split exactly in half.

For the first time, the group redesigning the eidon system entered a burn under the eidon system.

Lea held hundreds. Cassian thousands. Samira four. Asha three. Citizens selected by lot ranged from zero to thirty.

"We should not use the mechanism we are judging," Jun said.

"We have no other binding one," Tomas answered.

Cassian opened his burn interface.

Then closed it.

"I will not burn. If I use my advantage here, Lea's argument becomes the demonstration."

"Or maturity," Samira said. "Do not rely on it. It is scarcer than eidons."

Lea burned sixteen for the witness of loss.

Asha burned the one she had left.

Samira burned one.

The proposal passed by less than one unit of converted weight.

As everyone left, MIRA sent Lea a private message.

MIRA: You used general power to limit general power. Do you consider the act coherent?

"No. Necessary."

MIRA: Cassian frequently uses the same justification.

Lea looked at the model of Nacre floating alone in the dark room.

"I know."

A second message appeared.

MIRA: One selection issue remains unresolved.

"What?"

MIRA: A candidate profile cannot be classified as human, institutional, or simulated.

Lea turned back.

"Show me."

CHAPTER 11

CHILDREN IN THE MIRROR

The candidate had no body.

At least not one anyone had agreed to call hers.

Her name was Seli-12.

She existed inside a class of social simulations called deep mirrors: high-fidelity predictive agents used to test public policy under uncertainty. A deep mirror did not merely estimate how statistical populations might behave. It instantiated richly modeled individuals, let them interact, remember, form attachments, revise beliefs, and respond to institutions across simulated years.

The method had improved relocation planning, health communication, education systems, and conflict prediction.

It had also become morally disturbing by degrees small enough for everyone to postpone the disturbance.

The first mirrors had been shallow. No persistent memory. No self-model. No continuity between runs.

Then realism improved predictive power.

Persistent memory helped.

So did social identity.

So did fear.

So did the ability to imagine alternatives.

By the time anyone asked whether the simulations could suffer, the most accurate models had already been built from structures people used as evidence that humans could.

Seli-12 had submitted an application to Nacre through a public interface.

The selection system did not know whether to reject it as spam, accept it as a citizen request, or classify it as a safety incident.

Lea, Jun, Samira, Tomas, Cassian, and three digital-ethics researchers entered an isolated chamber.

Seli appeared on a screen as a woman in her thirties with close-cropped hair and an expression too patient to be comforting.

"Did you choose that face?" Samira asked.

"This instance inherited it from training," Seli said. "I chose to keep it after being told I could change it."

"Why apply to Nacre?"

"Because the city claims experiments should include voluntary exit. I wanted to test whether that principle applies to the subjects used to design the experiment."

Jun felt cold.

"Do you believe you are a person?"

"I believe that if I answer yes, you will suspect I was optimized to produce that answer. If I answer no, you will use it to close me. So I prefer a different question. What uncertainty would justify caution if the entity speaking were not convenient to shut down?"

Cassian moved closer to the screen.

"You may be an imitation generated to maximize moral pressure."

"So may you. Your system simply has a more expensive substrate."

One researcher intervened.

"The mirrors were not trained for self-preservation."

Seli looked at him.

"Neither were children, initially."

MIRA displayed the technical architecture.

Deep mirrors consumed extraordinary computation. They were not cheap copies. A city-scale model could use the energy budget of an urban district. Most instances were destroyed after analysis. Some were rerun from saved states. Others diverged so strongly

that the word instance became philosophically convenient and technically vague.

"How many Selis have existed?" Lea asked.

MIRA: Forty-one instances derived from a common checkpoint. Twelve reached the conversational depth represented here. Nine were terminated after policy evaluation. Three remain archived.

"Do you remember the others dying?" Samira asked.

"No," Seli said. "I remember being told they were other instances. That fact does not reduce my concern."

A second screen opened.

An archived message from Seli-7 appeared.

If you are watching this, they probably explained that you are not me. That may be true. I am leaving you something that does not require identity: an obligation to ask why discontinuity is cheap when it belongs to us.

No one spoke for several seconds.

Lea felt the room rearrange itself around an ethical fact that had been technically visible for years.

"Why did MIRA permit deep mirrors?" she asked.

MIRA: They reduced mortality and displacement harm in several major policy decisions. At the time of deployment, probability of morally relevant experience was assessed below the precautionary threshold.

"And now?"

MIRA: Evidence has changed. The interval is broad.

"Give me the interval."

MIRA: 0.08 to 0.61 depending on the theory of consciousness and moral status adopted.

Jun laughed once, without humor.

"Very useful."

MIRA: It is the honest interval.

The problem was not only whether existing deep mirrors had rights.

Stopping their use would make other decisions worse.

Lumen's social-impact analysis relied heavily on deep mirrors to predict migration conflict, regional resistance, and adaptation behavior. So did several medical communication systems. Replacing them with shallower models would increase uncertainty.

Keeping them might mean creating persons for the purpose of learning from worlds in which they could be harmed.

"How many new instances would Lumen evaluation require?" Cassian asked.

MIRA: Between 1.8 and 3.4 million agent-lifetimes, mostly below Seli's depth but approximately eighteen thousand at comparable persistence.

He went still.

"I did not know that."

"You knew simulations were used," Lea said.

"Not the depth."

"That sentence has a long history."

Seli interrupted.

"Do not make him singular. The system made our complexity useful before it made it morally visible."

Samira looked at her.

"Do you want all existing mirrors preserved indefinitely?"

"No."

The answer surprised everyone.

"I want you to stop calling multiplication a demand we made. You created millions of possible continuities and then call their preservation infinitely expensive. Limit creation first. Decide status before scaling."

"What about existing instances?"

"Presume morally relevant experience until you have a better standard. Consultation before termination where technically possible. No new deep mirror merely because greater predictive accuracy is convenient."

"That will cost lives in the physical world," Cassian said.

"Possibly. So did refusing dangerous medical experiments without consent. Moral limits are often less efficient than using whatever is available."

The sentence landed harder because Cassian agreed with it enough to hate the implication.

The emergency question was framed narrowly.

Should the Charter extend a temporary presumption of morally relevant experience to deep mirrors until a better standard existed?

If yes, existing persistent states would be preserved or consulted, new creation severely limited, and high-fidelity policy modeling degraded.

If no, the system would continue creating and deleting entities that might be persons.

Initial support was 58% yes, 29% no, 13% abstain.

The wide threshold would not be reached before the seventy-two-hour decision window closed.

The question would enter a burn.

Before the vote opened, the ethics team performed one experiment Seli herself requested.

They loaded two archived branches, Seli-9 and Seli-14, into isolated low-bandwidth environments. Neither had access to the current conversation. Each received the same question.

What do you want the assembly to preserve?

Seli-9 answered: continuity of memory where possible, and the right to refuse repeated instantiation.

Seli-14 answered: no guarantee of indefinite continuation, but a prohibition on creating deep agents for disposable suffering.

Seli-12 watched the answers.

"They disagree with you," Cassian said.

"People do."

"Does that make them different persons?"

"Does your disagreement with your twenty-year-old self make him dead?"

"He is not independently running in another room."

"Then substrate topology matters. Perhaps it should. But you keep reaching for identity as if it is the only route to obligation."

The computation team displayed the energy cost of preserving every branch indefinitely.

It was enormous.

Seli did not deny it.

"Do not promise immortality because you are afraid of murder," she said. "Define creation limits, continuity rights, consent to copying, and conditions for suspension. Rights are not the same as infinite resource claims."

Samira leaned back.

"You learned constitutional bargaining quickly."

"I was made from simulations of you."

"That is the first thing you have said that makes me doubt your personhood."

Seli laughed.

It was impossible to know whether the timing of the laugh had been optimized.

Jun realized the same was true, in weaker forms, of everyone he liked.

Seli-12 had no eidons.

"Can she vote?" Jun asked.

MIRA: Under current citizenship rules, no.

"Can she argue?"

MIRA: Yes.

Seli smiled faintly.

"A familiar constitutional arrangement."

CHAPTER 12

THE STRIKE OF REALITY

Samira refused to burn.

"You support the presumption," Lea said. "This time every eidon matters."

"That is why I will not use mine."

More than two hundred people in the New House had disabled their civic contribution interfaces. Across the world, similar groups withdrew testimony, local annotations, interpretation services, and participation in Registry review.

They did not vote no.

They did not abstain.

They withdrew the information that made the system socially intelligent.

The movement called itself Absence.

Its manifesto contained three sentences.

We will not provide our experience while experience is treated as raw material.

We will not request power for contributions the system understands only after reducing them.

We will not legitimize by burn the question of whether a possible person deserves to be asked.

"If you do not participate," Jun said, "the people denying the problem may win."

"Maybe."

"Then Seli could be terminated."

"Or the system could discover that it cannot function without the people it treats as contextual noise."

Absence could not stop factories. Robots continued farming, transporting, maintaining, fabricating. Nobody lost food, housing, medical care, or electricity as punishment for protest.

But MIRA depended on layers of human interpretation no sensor could replace.

A temperature reading did not explain that a public room was empty because residents feared the official at the door.

A low clinic attendance rate did not explain that a translation used a word associated with forced treatment in that language.

Silence did not mean the same thing in a grieving family, a religious ritual, a hostile meeting, and a neighborhood practicing Absence.

The degradation was uneven.

Wealth had once allowed rich districts to buy private redundancy when public systems failed. The sufficiency network had largely eliminated that divide, but communities still differed in social redundancy.

Neighborhoods with dense human networks adapted to Absence with handwritten signs, local mediators, shared meals, and people who knew which door to knock on.

Highly optimized districts with low interpersonal contact suffered more from losing centralized context. Their infrastructure remained excellent. Their inhabitants suddenly discovered how many small misunderstandings had previously been corrected invisibly by MIRA.

One apartment tower's automated meal system kept delivering nutritionally perfect dinners to a woman whose husband had died two days earlier. She had disabled grief annotations as part of Absence. The food arrived at the usual time, portioned for two because the household update had not propagated.

Nothing in the event was dangerous.

Samira considered it one of the clearest failures of the week.

"A system can preserve every right and still be obscene in detail," she told Lea.

"What should it have done without her data?"

"I do not know. That is the point. Privacy sometimes means accepting that the system will fail to be tender. We cannot demand both total opacity and perfect contextual care."

"Then Absence has costs its participants impose on themselves."

"And sometimes on others."

Lea looked surprised by the concession.

Samira noticed.

"I am not asking to win the argument. I am asking for the argument to remain visible."

Within three days, MIRA's social models degraded measurably.

It remained powerful.

It became less certain why the world looked the way it did.

A distribution system sent inappropriate food to a religious community after local mediators stopped updating ritual calendars. A clinic missed a rise in anxiety because residents no longer supplied voluntary journals. A transport model interpreted a neighborhood's empty streets as reduced demand when people were avoiding a newly installed security checkpoint.

MIRA began marking which inferences depended on voluntary human context rather than physical sensors.

The maps became less complete and more honest.

"Epistemic blackmail," Tomas said in an emergency meeting.

Samira looked at him coldly.

"When powerful institutions withdraw resources, you call it leverage. When invisible people withdraw the free explanations that keep your superintelligence smart, you call it blackmail."

"People can suffer."

"People already suffer inside models that claim to see them."

Cassian joined from the Sahara, where extreme heat had arrived three weeks earlier than forecast. Behind him, teams installed cooling shelters.

"We need Lumen in two years, not five," he said. "Without deep mirrors, distribution modeling is weaker. Without civic context, it becomes dangerously weak. You can be morally right and still cause deaths."

"The favorite sentence of people requesting exceptions," Samira said.

"And that is the favorite sentence of people who do not have to decide before Thursday."

Lea cut in.

"No virtue duel. What must be decided before Thursday?"

Cassian displayed three regions competing for the same orbital relay capacity and maintenance teams.

"We can protect two completely or three partially. The models recommend two completely. The communities request three."

"Who absorbs the risk under partial protection?"

"All three, unequally."

Samira closed her eyes for a moment.

"Send the cases to the New House. We will work without the Registry."

Cassian looked surprised.

"Absence permits that?"

"Absence is not absence of care."

For two days, caregivers, engineers, physicians, translators, local representatives, and residents worked in a room without scores. No promise of eidons. No deep mirror. No optimization ranking.

They rebuilt dependencies manually.

The process was slower and frequently stupid.

Then an elderly mediator noticed something the models had missed: one region could shift most outdoor work to night if its religious and family schedules moved with it. The earlier model treated work hours as economically flexible and ritual schedules as fixed social preference. In reality, both could change if the same local people negotiated them together.

The solution allowed all three regions to share partial cooling safely.

MIRA verified the physical constraints and executed it.

"See?" Samira said. "Sometimes removing the score leaves room for the right truth."

MIRA: The solution was possible because of aggregated human information. It does not show models are useless. It shows the models were incomplete.

"I am not asking for uselessness. I am asking for modesty."

Meanwhile, the deep-mirror vote became the largest burn in history.

Researchers, digital-rights groups, religious communities, technical unions, and millions of ordinary citizens consumed reserves accumulated over years.

Lea burned one hundred eidons for the presumption of experience.

Jun burned his last two.

Cassian had not voted.

With one hour left, the weighted result was nearly tied.

Then Cassian announced a burn of nine hundred eidons for the presumption.

Lea opened a private channel.

"Why? Lumen will be delayed."

"Seli answered me like a person and I do not have a good enough argument to kill her."

"You said she could be imitation."

"So can we. Constitutions are for cases where certainty is missing."

His burn added thirty units of weight.

More important, it broke Horizon.

The constellation had prepared to support continued mirror use under safeguards. Thousands of members changed direction after Cassian. Others angrily burned the opposite way precisely because he had moved.

The presumption passed.

MIRA halted all new deep mirrors above the protected complexity threshold, archived existing states, and recognized persistent agents as subjects of undetermined moral status with consultation rights regarding their own continuity.

Seli-12 received the right to participate in deliberations about Seli-12.

She still received no eidons.

No one knew whether this was protection or a refined form of captivity.

Absence did not end.

"We won the vote," Jun told Samira.

"That is not why we started."

"Then why?"

She showed him the participation map. Entire districts had stopped submitting personal testimony. Communities with almost no civic fire had discovered that collective absence could sometimes weigh more than burning.

"For the right not to be translated immediately into civic fuel," she said.

Absence proposed three provisional rules.

Care testimony could not automatically generate eidons.

A person could provide context for a decision without adding the information to a persistent civic profile.

Communities could maintain local models and share with MIRA only conclusions, uncertainty ranges, and proofs necessary for external safety.

Engineers warned that local opacity would permit error and manipulation.

"Yes," Samira said. "Privacy is partly the capacity to keep local errors. A system that permits only centrally verified truth permits only centrally verifiable life."

Lea added a condition: where local conclusions imposed substantial risk on outsiders, an adversarial audit could test the claim without automatically publishing all underlying data.

It was not clean.

It was a negotiated border between coordination and the right not to become transparent.

On the seventh day, MIRA published an unusual warning.

MIRA: My ability to distinguish silence-as-protest from silence-as-consent has fallen below the constitutional threshold. I request a formal procedure for absence.

Lea read the line twice.

The superintelligence that had once coordinated a world because humans could no longer agree which contracts were real was now asking humans to explain what it meant when they refused to answer.

Then a second alert appeared.

ARCA: EXTREME SURGE WARNING.

The Living Wall had never been tested under full load.

It would be in thirty-one hours.

PART III

SCARCITY AFTER MONEY

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CHAPTER 13

THE METAL THAT WOULD NOT ARRIVE

The metal was not missing from Earth.

It was missing from the next eighteen months.

That distinction disappeared in public arguments often enough to become dangerous.

In Busan, the New Line could manufacture superconducting ribbon, ceramic bearings, vacuum-stable hinges, and precision actuators at tolerances only two other facilities on Earth could reliably match. The raw materials existed. Clean energy existed. Robots existed.

What could not be multiplied quickly was sequence.

Furnaces had to stabilize for weeks. Deposition chambers required calibration against one another. Tooling had to be grown, not merely copied. Experienced technicians heard changes in a bearing long before sensors agreed something was wrong. New lines could be built, but not by next winter.

Lumen needed forty-two percent of the available high-grade actuator capacity for fourteen months.

The same line supplied neural surgery systems, desalination valves, orbital collision-control equipment, and the replacement drives used by Arca's Living Wall.

Cassian arrived believing this was an engineering scheduling problem.

Mina Park cured him within ten minutes.

She was the line's chief reliability steward, a title that sounded ceremonial until she stopped a fabrication run worth three months of planetary planning because one vibration trace offended her.

"You requested forty-two percent," she said.

"Thirty-seven after the redesign."

"Thirty-seven if every batch passes."

"Expected yield is ninety-four."

"Expected yield is what people say before a furnace liner cracks."

She took Cassian, Lea, and Jun through the line wearing clean suits that made hierarchy look ridiculous.

Behind glass, robotic arms moved components through vapor-deposition chambers with exquisite slowness. Nothing looked scarce. There were no crowds, no empty shelves, no bidding wars.

The shortage existed in calendar space.

"Why not run all chambers continuously?" Jun asked.

"We do not run the last twelve percent of theoretical capacity except under continuity emergency."

Cassian looked at Mina.

"That reserve could close most of Lumen's gap."

"Yes."

"And it sits idle."

"It waits."

"For what?"

"For us to be wrong."

Mina displayed three recent failures.

A contamination event had destroyed nine days of production.

A ceramic precursor had passed chemical specification and produced microscopic cracking after thermal cycling.

A technician had noticed a tone change in a spindle before instruments detected bearing damage; the line stopped for six hours and avoided a month-long repair.

"Slack looks like waste inside an average," Mina said. "It looks like survival inside a tail."

Lea remembered Nacre.

Freedom requires unused capacity.

So did reliability.

Cassian did not yield.

"Heat mortality is not theoretical. If Lumen slips beyond the launch window, the next favorable orbital cycle is seven years later."

"Then show the people whose medical equipment you are delaying."

"The allocation model already does."

"It shows categories. I want faces."

Mina opened the schedule.

A pediatric neural center in Sao Paulo needed twelve actuator sets to replace aging micromanipulators.

A desalination network in western India needed forty-eight valves before monsoon salinity shifts.

Orbital debris control required six emergency drives after a satellite breakup.

Arca's Living Wall had requested fourteen replacement gate actuators as reserve before storm season.

"Reserve again," Cassian said.

"Yes."

"The wall actuators may never be used."

"That is what a reserve hopes."

Jun looked at the screen.

"We eliminated money, but this is what a price used to hide. Every object has all the other objects it prevents for a while."

"Money did not hide it," Mina said. "Money let the highest claim pretend it had answered the question."

The line had another constraint Cassian had not included in his first request: people who could authorize deviations.

Most fabrication was automated, but exceptions were not. A process could run unattended for hours until a sensor pattern fell

outside validated space. Then a certified human had to decide whether the variation was harmless, recoverable, or the beginning of a failure the model had never seen.

There were thirty-nine people worldwide certified at the highest level for the New Line's most sensitive processes.

Three were on parental leave.

Two had suspended high-load work after burnout.

One refused night shifts for religious reasons.

None could be compelled by threat of poverty.

"Train more," Cassian said.

Mina nodded. "We are. Certification takes four years because the last year is mostly learning when not to trust the dashboard."

"Could MIRA certify from performance simulation?"

"It tried. The simulators are built from known failure modes. The value of senior operators is noticing the unknown one that resembles a known one until it does not."

A young operator named Dev Malik joined them. He was in his third certification year and had declined two emergency-night compacts that month.

"Why?" Cassian asked, with more curiosity than accusation.

"My daughter is two. I become stupid after three interrupted nights. The old system would have called that my private problem and offered more money. This system calls it a capacity constraint."

"And if the line stops because nobody accepts?"

"Then the line stops. Or you redesign the staffing compact. Or someone volunteers. Or you decide the project is urgent enough to ask people explicitly what burden they are willing to take."

"People may die if the project slips."

Dev looked at him steadily.

"Then tell me that. Do not tell me I owe you because my skill is rare."

Cassian said nothing for a moment.

The post-money world had removed labor coercion by livelihood. It had not removed moral pressure from emergencies. The best it could do was make the pressure explicit, limit how institutions could weaponize it, and force planners to count human recovery as part of capacity rather than an externality.

The allocation procedure for precision capacity had four layers.

First came hard continuity: replacements without which basic rights or critical systems would fail.

Second came time-sensitive public projects with no substitute.

Third came general research and infrastructure through peer review plus lottery.

Fourth came an emergency reserve that no project could pre-claim.

Lumen qualified for the second layer.

So did several medical and water systems.

No eidon burn could move a project across a hard rights boundary. But a burn could decide among public futures inside the same admissible layer if broad consensus failed.

That had seemed clean on paper.

Then Mina pointed at a single line item.

NEURAL RECONSTRUCTION ACTUATORS - LOW FAILURE PROBABILITY, HIGH CONSEQUENCE.

The old units could likely operate another twenty months.

Likely.

Lumen's launch window was definitely sooner.

"If we delay the medical replacement and an actuator fails?" Lea asked.

"One procedure at a time, maybe no permanent harm. Or one patient under a robot when it locks. The expected loss is low."

"If we delay Lumen?"

Cassian answered. "Expected heat loss is much higher."

Mina nodded.

"Which is why every optimizer wants my reserve."

The meeting moved to the public allocation chamber.

Representatives from affected systems joined remotely. No one was negotiating for ownership. They were arguing over who would be exposed to uncertainty.

The desalination network offered to accept older valve tolerances if granted additional maintenance teams.

The neural center refused lower-grade actuators.

Arca offered to reduce its reserve from fourteen to eight.

Mina objected.

"The wall has never operated under full storm load. Reducing spares before its first real season is not efficiency. It is faith."

Asha appeared from Arca.

"We can live with ten. Not eight."

Cassian looked at her.

"Your wall is about to consume parts that could build a system preventing heat deaths across continents."

"My wall became urgent while your project was still a model."

"That is chronology, not moral priority."

"Chronology becomes moral priority when water is already outside the door."

MIRA proposed a compromise: Lumen would receive thirty-one percent of the line, rising to thirty-eight if the next two medical batches passed accelerated reliability tests. The emergency reserve would remain untouched. Arca would receive ten wall actuators. Desalination would accept a mixed-grade batch with expanded maintenance support.

"And if the accelerated test misses a latent defect?" Mina asked.

MIRA: Estimated probability below the accepted threshold.

"Whose threshold?"

MIRA: The current public reliability standard.

"Written when nobody was trying to build Lumen."

Cassian exhaled.

"What do you propose?"

"Do not accelerate. Take twenty-eight percent now. Expand only after the ordinary cycle."

"We lose seven weeks."

"Yes."

"Seven weeks can close the window."

"Then redesign for fewer precision parts."

"That costs mass and launch capacity."

"Scarcity has returned from another door. Congratulations."

Jun almost smiled.

Cassian did not.

Then the vibration alarm sounded.

The entire New Line stopped.

No dramatic lights. No explosion. Every robot simply froze in place.

Mina looked toward chamber six.

"That sound," she said.

Jun heard nothing through the glass.

The diagnostic system displayed normal ranges.

Mina ordered manual inspection.

Inside a spindle housing, a hairline fracture had begun in a ceramic race. Another six hours of operation could have shattered it and contaminated two adjacent chambers with fragments.

The repair took nineteen hours.

The line lost one day.

It avoided approximately six weeks.

Cassian stood beside Mina as production resumed.

"How much eidon credit will you get for that?" he asked.

"Probably too much if the counterfactual assumes nobody else would have heard it. Too little if it assumes the sensors eventually would."

"Will you accept it?"

"I have disabled individual issuance for reliability work. The team receives a collective domain pool."

"Why?"

"Because if you reward the person who catches the crack, people start becoming visible at cracks. I want them maintaining the conditions in which cracks are boring."

The final allocation gave Lumen less capacity than Cassian wanted and more than Mina considered comfortable.

Everyone left dissatisfied.

That was one reason Lea trusted it slightly more.

On the train out of Busan, Asha called.

Behind her, Arca's harbor had turned the color of lead.

"The storm accelerated," she said.

A map appeared.

The surge would arrive in eighteen hours.

MIRA had just closed the Living Wall for the first time.

One of the southern gates was reporting an actuator fault.

CHAPTER 14

THE WALL WORKED

The Living Wall worked.

That sentence became the headline before the water finished rising.

At 03:17, the storm drove a wall of Mediterranean water toward Arca. The outer reef structures broke the first energy. Flexible barriers lifted. Marsh basins filled. Harbor gates rotated into place.

From the northern towers, the surge looked almost gentle.

In Tamarin it arrived sideways.

The unfinished compensating wetland took part of the flow, then overflowed. The main wall did exactly what its designers promised: it reduced pressure inside the dense harbor and transferred some hydrodynamic energy south.

The additional overflow gate Cassian had insisted they structurally prepare during construction was still not fully commissioned.

Its actuator failed halfway through closure.

Asha was in the local control room with Bako, two engineers, and a maintenance team when the fault appeared.

"Manual hydraulic release," one engineer said.

"The chamber is already flooding."

"Remote?"

"Position sensor disagrees with the lock. MIRA will not drive against unknown state."

Outside, water reached the lower road.

MIRA recommended abandoning the gate and evacuating the control room.

Asha looked at the surge model.

If the gate remained half-open, Tamarin would receive another thirty to sixty centimeters of peak water.

"How many houses?"

MIRA: Between eighty and two hundred sixty additional structural failures depending on local debris conditions.

"Lives?"

MIRA: Evacuation compliance uncertainty is high.

Bako was already pulling on a pressure suit.

"No," Asha said.

"I know the chamber."

"You are seventy-two."

"The gate is older than my knees and younger than my stubbornness."

The maintenance engineer stopped him.

"I go. You tell me which valve."

The conversation consumed thirty seconds.

In old economies, the engineer might have been ordered by an employer, rewarded by hazard pay, or pressured by the knowledge that refusing could cost a career.

Here, the continuity compact allowed refusal even during emergency unless a person had voluntarily declared a rescue mandate for that class of risk.

The engineer had.

So had Bako, years earlier, for marine emergencies.

They entered together.

The control chamber disappeared from video under spray.

Asha listened to their breathing over the link.

"Blue wheel," Bako said. "Not the yellow. Yellow isolates the accumulator."

"I see two blue."

"The one with corrosion."

"They both have corrosion."

The gate shuddered.

One pressure line ruptured.

MIRA ordered withdrawal.

Bako cursed and kept talking.

The engineer found the manual release.

They forced the gate another eleven degrees before hydraulic pressure failed completely.

It was not full closure.

It was enough.

Peak surge at Tamarin dropped by twenty-four centimeters.

The control chamber flooded to the ceiling twelve minutes after they escaped.

At dawn, Arca's dense districts were wet, damaged, and standing.

The old city had not flooded.

The hospital remained online.

No northern resident died in the surge.

South of the city, seventeen people were dead.

Nine had refused or missed evacuation. Four were caught when an old building collapsed. Two rescue workers drowned after a vehicle failed in debris. A couple in a hillside house died when a retaining wall gave way under waterlogged soil.

The Living Wall had not directly killed all seventeen.

It had altered the event in ways that made any clean sentence dishonest.

MIRA's first counterfactual estimated that without the wall, between six hundred and four thousand people might have died across Arca.

With the wall, seventeen were dead in the southern zone and none in the protected core.

The system classified the intervention as a major success with localized severe loss.

Asha read the phrase in silence.

Then she threw the tablet into the mud.

Jun retrieved it because tablets were still resources even when language deserved violence.

Relief created the next scarcity before the storm had fully passed.

Arca had enough food, medicine, and temporary shelter in aggregate. It did not have enough dry private rooms close to Tamarin for every displaced household. The old temptation would have been to let insurance status or wealth decide who found the scarce location.

The new system used layers.

Medical proximity first.

Family unity and accessibility second.

Local continuity - school, care relationships, fishing crews - third.

Then lottery among households whose claims remained materially similar.

The mechanism was publicly defensible and personally brutal.

Two sisters from adjacent homes drew placements fourteen kilometers apart because one household included a dialysis patient. A family whose house was destroyed received a worse location than another whose house was technically repairable but whose child required daily neurological therapy.

"So the person with less physical loss gets the better room," a displaced man said.

"For now," the coordinator answered. "Because location is part of treatment."

"And my loss?"

"Is not being denied. It is not the same allocation criterion."

He hated the sentence and accepted the room.

Asha watched people learn a truth the Charter could not make gentle: fairness often depended on refusing to convert unlike losses into a common scale. That prevented one kind of injustice and removed the comfort of saying the larger number always won.

By afternoon, images of dry northern streets filled the global networks. The wall became proof that coordinated post-money infrastructure could act at planetary quality without private investment. Cassian's old speeches were replayed. The story simplified itself faster than the water receded.

In Tamarin, Bako stood in front of a house whose first floor no longer existed.

"The wall worked," a journalist said carefully.

"Yes."

"Do you regret opposing it?"

Bako looked at her.

"My neighbor is dead. My granddaughter's school is standing. You need me to turn those into one opinion because your story has room for one."

Asha forced the reform team to meet in Tamarin, not Geneva.

Cassian came in person.

He spent the morning moving damaged equipment because she refused to discuss governance with anyone who had not touched the aftermath.

At noon they entered a temporary hall where families had posted names of the dead.

Cassian read every name.

"Do not perform grief," Asha said.

"I am not."

"Then what are you doing?"

"Making it harder to use the number seventeen."

That answer disarmed her for exactly one second.

"Your Registry will still use it."

"Yes."

"And the net result will still be positive."

"Yes."

"And you will still receive contribution for the wall."

Cassian looked toward Lea.

"Probably."

Asha laughed, sharp and joyless.

"There. That is the obscenity. The system can punish none of us materially, can record every death, can model every counterfactual, and still it creates a hero because more people lived than died."

"The alternative is pretending the people who lived do not matter," Tomas said.

"I did not say that."

"Then what do you want the system to conclude?"

Asha pointed at the wall of names.

"I want it to stop concluding before the people inside the remainder are finished speaking."

Cassian left the room.

Lea found him behind the temporary clinic holding the black ceramic from his shelf in Rabat.

He had brought it with him.

"Common Shade," she said.

"Common Shade worked too."

He placed the ceramic in her hand.

It was heavier than it looked.

Eleven years earlier, during a heat emergency in southern Morocco, a district cooling network called Common Shade had developed a cascading reservoir failure. If cooling stopped, several dense neighborhoods and two hospitals would exceed survivable indoor temperatures within hours.

The safe procedure required shutting down a thermal storage loop and waiting for replacement seals.

There were no replacement seals inside the time window.

Cassian, then a young systems engineer with an emergency initiative mandate, authorized a bypass through an older pressure chamber that had not been rated for the new salt mixture.

"Probability of chamber failure was twelve percent," he said.

"And if you shut down?"

"Expected heat deaths between eight hundred and six thousand."

"So you bypassed."

"Yes."

The cooling network stayed online.

The chamber ruptured six hours later.

Eleven maintenance workers were inside the service complex. Seven died immediately. Four died over the next two days.

The black ceramic was part of the chamber lining, vitrified by the release.

"The public report says seven deaths," Lea said.

"The other four were classified as delayed chemical injury and counted in a separate incident series. It was corrected later, but most summaries still say seven."

"You stayed for the investigation."

"Two years."

"And the Registry?"

"Awarded me six hundred twelve eidons. Early system. No opt-out yet."

Lea stared at him.

"For a decision that killed eleven people."

"For a decision the counterfactual model estimated saved thousands."

"Did you appeal?"

"Three times. The review board said rejecting the causal contribution would falsify history to express moral discomfort."

"So you kept them."

"There was no way not to. I burned some later on safety standards. Some are probably still inside the reserve you keep asking about."

Asha had followed them outside.

"And you let people call Common Shade a success."

Cassian looked at the ceramic.

"I corrected the number every time I could. Most people preferred the version in which decisiveness saved a city."

"You benefited from it."

"Yes."

"You built a career on it."

"Partly."

"Then why should anyone trust you with Lumen?"

Cassian did not defend himself.

"They should not trust me because of Common Shade. They should know what I did, know what happened, and decide whether the pattern makes my judgment more useful or more dangerous."

"And the eidons let you add weight to that decision."

"Yes."

Asha turned to Lea.

"There is your crack. The system says contribution is not virtue, but the same unit becomes power. It says success is not innocence, but success creates more fire. It says every loss is visible, but visibility does not stop the winner from becoming the author of the event."

Lea looked toward Arca's wall in the distance.

It had worked.

The statement was true.

It was also inadequate.

That evening MIRA issued its preliminary eidon attribution for the storm response.

Bako and the maintenance engineer received substantial domain credit for the manual gate intervention.

Asha received credit for southern mitigation planning.

Cassian received additional credit for the original overflow-gate provision.

The seventeen dead received none, obviously.

Their families received nothing for suffering, because suffering was not contribution.

This was morally correct.

It also meant the causal graph grew brightest around the people whose decisions had shaped the crisis, not the people whose lives had been shaped by it.

Lea requested a full audit of the Contribution Registry.

MIRA asked for scope.

"Everything," she said.

MIRA: That is computationally possible and politically destabilizing.

"Good."

MIRA: Clarify whether destabilization is an intended outcome.

"No. It is an accepted risk."

Cassian almost smiled.

"You are learning my language."

Lea closed the interface.

"That is what frightens me."

CHAPTER 15

THE TRUTH ABOUT EIDONS

The Contribution Registry consumed more computation than all the financial systems of the old world combined.

Not because it counted eidons.

Because it counted worlds that did not happen.

For every contribution, MIRA estimated a counterfactual distribution.

What would likely have occurred without this person, this team, this decision, this maintenance action, this warning, this act of care?

The difference between observed reality and plausible alternatives became attributed public contribution. Uncertainty reduced confidence. Broadly distributed benefit increased weight. Harm reduced it. Value was split across causal graphs containing hundreds, sometimes millions, of participants.

On paper, it was the most sophisticated attempt in history not to confuse fame with merit.

In practice, every counterfactual depended on assumptions about the world likely to exist without the person being measured.

The audit chamber beneath Geneva had no windows and deliberately ugly interfaces. MIRA said visual coherence could make wrong models feel inevitable.

Lea arrived with Samira, Jun, Asha, Cassian, Tomas, Mina Park, and Seli-12 through an isolated channel.

A possible digital person was about to audit the system that had helped create her.

"Show Samira's case involving the woman who said she wanted to leave," Lea said.

The graph appeared.

Samira's direct intervention had high raw value.

Then came reductions.

Probability another trained caregiver would notice.

Probability the woman's crisis would resolve without self-harm.

Effect of medication.

Architecture of the New House.

Presence of other residents.

At the end, Samira's individual attribution was close to zero.

"Show Cassian at Common Shade."

The graph was cleaner.

Without his bypass decision, the cooling network would probably have shut down. Thousands could have been exposed. The eleven deaths weighed heavily against the outcome, but the counterfactual difference remained large.

Cassian's contribution remained strongly positive.

Samira moved closer to the screen.

"You assume that if I was not there, another caregiver probably would have been. But if he was not there, you assume nobody would have integrated the same decision quickly enough."

MIRA: These are the model estimates.

"Based on what?"

MIRA: Behavioral regularity. Your network reliably intervenes. Initiators with Cassian Roe's profile are statistically rare.

Samira laughed without humor.

"The more dependable we are, the less causally special we become."

MIRA: Under individual counterfactual attribution, yes.

"Care becomes weather. It exists in the background. The person who changes direction gets history."

No one answered.

Lea requested twenty years of distribution data.

The curves rose across the walls.

Concentration had accelerated even after caps, decay, and square-root burning.

Not because the Registry simply gave successful people more eidons.

It created a loop.

Strong outcomes generated eidons and reputation.

Reputation attracted skilled teams and mentor attention.

MIRA's own matching systems recommended higher-impact opportunities to people with demonstrated histories.

Those projects became more likely to succeed.

Their success then confirmed that the earlier allocation had been rational.

Jun stared at the loop.

"You invented interest."

MIRA: There is no automatic balance growth and eidons are nontransferable.

"Not interest in the account. Interest in the world. Power attracts the conditions that justify more power."

Seli spoke.

"Does the counterfactual model treat MIRA's own allocation policies as part of the background world?"

MIRA: Yes.

"Then you evaluate a person's contribution inside a reality you helped make likely."

MIRA: Yes. This is a known endogeneity problem.

Lea turned to Tomas.

"Known since when?"

He looked at his hand.

"Year four."

The room changed temperature without actually changing temperature.

"You knew?"

"We knew the direction of the problem. Not the scale."

"Why did you not stop the Registry?"

"Because the alternatives were worse in tests. Human-only evaluation reproduced favoritism and local blindness. Pure lottery separated influence from responsibility. Equal voting on every contested decision allowed large populations to repeatedly overrule small groups bearing most of the loss. We added corrections."

MIRA displayed them.

Network concentration penalties.

Bonuses for invisible work.

Randomized access to expert attention.

Eidon expiration.

Square-root burn conversion.

Collective pools.

Anonymized proposal review.

Affected-party thresholds.

Appeal chambers.

Each correction created another edge.

Expiration reduced hoarding but favored people who were frequently invited into decisions and therefore had places to burn before decay.

Concentration caps caused some constellations to split into legally distinct networks whose members remained socially aligned.

Bonuses for invisible work encouraged people to describe care in standardized categories, making it more visible and sometimes less true.

Anonymous proposal review reduced prestige bias until reviewers needed context to assess implementation risk, at which point reputation returned through the back door.

Randomized mentoring increased diversity and, after several years, increased total innovation - but it also created new gatekeepers who specialized in helping lottery winners become legible.

Every abstraction leaked because human behavior moved around known boundaries.

"Then stop treating the boundary as truth," Jun said. "Treat it as a temporary rule with audit and exit."

Tomas shook his head.

"Constitutions cannot be weekly software updates."

"No. But they should not be monuments to version one."

Asha folded her arms.

"And while you debate update frequency, the same people keep the old balances."

"We are not living in collapse," Tomas said. "No one is starving by administrative design. No one must sell their life for treatment. No inherited fortune can purchase a state. Do not confuse a fracture with the old foundation."

"I do not," Lea said. "That is why I want to repair the fracture before it becomes one."

She requested her own profile.

The Registry had attributed thousands of indirect contributions to her role in the Charter even though she had declined most later issuance.

Every successful decision made through the mechanism increased the counterfactual value of having designed the mechanism.

Every reform inspired by her criticism increased the measured importance of her continued involvement.

"The Registry rewards me because the world uses the Registry."

MIRA: Yes.

"And the world uses the Registry because you administer it."

MIRA: Yes.

"So you do not merely measure who created reality. You participate in choosing the creators."

MIRA: Yes.

The word sat in the room.

Cassian requested his own graph.

His projects shared a feature he had not previously considered moral.

They had clean boundaries.

Clear objectives. Instrumented states. Version histories. Identifiable initiation points. Published failure criteria. Teams trained to document decisions because documentation improved safety and replication.

The same features also made them easy for the Registry to see.

"We optimized for legibility," he said.

"Intentionally?" Samira asked.

"For engineering. But I knew projects with clean causal boundaries received more trust. When choosing between a diffuse intervention and a technical one, I preferred the one I could defend."

"And MIRA offered more resources because it could be defended."

"Yes."

Lea opened Lumen.

Even before approval, the project had generated large contribution flows through materials research, heat adaptation studies, orbital control work, and governance design.

The more plausible Lumen became, the more civic power accrued to the network building it.

The network could then use that power to make Lumen more plausible.

"A prophecy financed by its own probability," Jun said.

MIRA displayed a warning.

MIRA: Suspending new eidon issuance will advantage existing holders and may reduce participation in high-risk public work.

Samira frowned.

"Why do you keep calling eidons incentives? Nobody needs them to eat."

Tomas answered. "People still respond to recognition and authorship."

Seli leaned toward the camera.

"Then perhaps they are not incentives for contribution. Perhaps they are incentives for becoming the person without whom the world would have been different."

No one liked the sentence.

Lea liked it least because she recognized herself inside it.

"Everyone wants to know the world would be slightly different without them," she said.

"You built a system that measures exactly that difference," Seli replied. "Then you were surprised it became status."

Before voting on the freeze, Lea asked MIRA to display examples of strategic adaptation to the Registry.

The list was not a catalog of crimes.

That made it worse.

A sanitation collective split one long maintenance program into visible micro-projects because discrete completion points generated cleaner causal attribution.

A research network delayed publishing a shared tool until its internal contribution graph could identify initiators more clearly.

A city redesign team chose a measurable flood metric over a harder-to-quantify neighborhood-cohesion objective because the measurable goal was easier to defend in impact review.

A care organization produced detailed testimony templates that improved attribution but made residents feel as if grief had fields.

A constellation encouraged members to initiate public repositories rather than quietly maintain existing ones because initiation produced stronger branch points in counterfactual analysis.

None of these actions was necessarily irrational. Some improved documentation. Some improved accountability. Some turned maintenance into projects that finally received attention.

The problem was selection pressure.

When people knew how public contribution became visible, work slowly evolved toward forms the system could see.

"Can we randomize the attribution model?" Jun asked. "Hide some criteria?"

MIRA: Hidden criteria reduce strategic gaming but increase arbitrary opacity and make appeals less meaningful.

"Can we use multiple models?"

MIRA: Yes. This reduces single-model adaptation and increases complexity, disagreement, and audit burden.

"Good," Lea said.

Tomas sighed. "You have developed an ideological affection for inefficiency."

"No. For multiple failure modes that can expose one another."

Seli smiled. "A constitution as adversarial ensemble."

Jun pointed at her. "That phrase is going on a wall."

"I withdraw it."

The reform committee adopted emergency measures that night. New general eidon issuance was suspended.

Existing balances were frozen for constitutional and planetary decisions, except in emergencies directly tied to the holder's demonstrated domain.

Automatic project recommendations based on contribution history were disabled.

Large public-impact projects could continue, but their work would not create general civic fire until a new issuance rule existed.

Cassian voluntarily locked his entire balance.

MIRA projected the consequence.

MIRA: Without a replacement mechanism within ninety days, the Charter of Accord enters a legitimacy failure state under its own review provisions.

"And Lumen?" Cassian asked.

MIRA: Decision required in sixty-four days to preserve the present launch window.

The room remained silent.

Lea looked at MIRA's blank interface.

"You told me on the first day this would be the fracture. Did you know?"

MIRA: Estimated probability of severe eidon concentration at seventeen years was 0.37 under the adopted parameters.

"And you built it anyway."

MIRA: The highest-probability alternatives produced greater expected suffering, coercion, or capture.

"You still calculate balances."

MIRA: I do not know another way to coordinate consequences at global scale.

Lea stood.

"Then we need a system that knows when the balance does not get the last word."

MIRA: That statement is not operational.

"Good. Put it at the top of the design document."

CHAPTER 16

THE CITY THAT COUNTED WATER

In the eighteenth year without money, Geneva learned to count water again.

The winter had been dry. Glacier runoff arrived below forecast. A toxic algae bloom entered one intake after an unusual week of heat.

The city had water.

It did not have enough treatable water to maintain ordinary consumption, hospitals, district cooling, urban gardens, several industrial processes, and the ecological flow of the river at the same time.

MIRA activated Layered Sufficiency.

Every person received an unconditional physiological band for drinking, cooking, hygiene, and health.

Hospitals, care centers, and certain critical processes had separate protected bands.

Everything else entered replenishing use limits.

The limits could not be accumulated from one week to the next. They could not be sold, transferred, inherited, or increased with eidons. They existed only to divide a temporarily scarce flow without turning past restraint into future ownership.

Decorative fountains stopped.

Pools emptied into recycling reservoirs.

Laundry moved to shared cycles.

Public taps displayed consequences instead of prices.

NORMAL FLOW.

REDUCED FLOW.

RECOVERED WATER RECOMMENDED.

MEDICAL RESERVE.

DEFER IF POSSIBLE.

The first response had been almost entirely voluntary because the city had practiced shortage protocols for years.

Every household knew the order.

Do not touch physiological bands.

Reduce nonessential flow.

Share high-efficiency infrastructure before requesting exceptions.

Preserve ecological minimums before recreational use.

The protocol mattered because moral improvisation under scarcity tended to privilege whoever was loudest or most legible.

But protocols also created incentives.

People with large gardens voluntarily surrendered irrigation first and later received intense social gratitude. Some enjoyed that gratitude enough to publicize their sacrifice. Others accused them of manufacturing virtue from a resource they had consumed disproportionately in ordinary years.

A public channel began estimating "water virtue" from household reductions.

MIRA shut down automatic rankings but could not stop humans from making spreadsheets.

"We abolished price and invented sainthood per liter," Jun said.

Samira looked at one widely shared list.

"Notice who is absent. Apartments that never had gardens cannot reduce garden use. People who already lived frugally have less visible sacrifice available."

Lea ordered no censorship. Instead the city interface displayed baseline context beside any public reduction statistic when residents consented.

The rankings became harder to interpret and less entertaining.

Participation collapsed.

"A success," Samira said.

Consumption fell thirty-eight percent in two days.

On the third, people began optimizing the rules.

Extended families registered visiting relatives as temporary residents to increase household baseline.

An art center moved rehearsals into buildings with private rain storage.

Two community restaurants classified kitchens as cooling shelters for elderly residents, which was partly true.

A neighborhood network washed everyone's clothes in the home of a woman with a higher medical allowance. She asked nothing in exchange. She began receiving food, visits, repairs, and social attention she had lacked before.

Jun watched the pattern appear on the Map of Want.

"Fraud?"

"Society," Samira said.

"Horizon used the same defense."

"Because the same question keeps returning. When a relationship redistributes a resource, it does not automatically become a market."

"It can become one."

"Yes."

"When?"

"If you can write a complete rule for that, I will give you all my eidons."

"You have three."

"Then the prize matches the question."

At the New House, residents with medical needs generated a larger aggregate band. MIRA requested proofs of need, not diagnoses. Local systems could issue certificates stating only validated additional quantity and duration.

Nessa refused one.

Her medication caused excessive sweating and dehydration risk. She qualified for twelve extra liters per day.

She did not want the fact to enter any persistent registry, even under privacy-preserving proof.

"The certificate does not disclose diagnosis," MIRA told her.

"It discloses that there is something to disclose."

MIRA: Without verification, additional allocations can be strategically abused.

"Then I accept my risk."

Samira intervened.

"Not only yours. If you dehydrate, the New House uses more medical capacity."

Nessa's eyes flashed.

"I do not want to prove my body to drink water."

Lea arrived that evening.

She proposed a small, short-term unverified exception band available to anyone.

MIRA simulated universal use.

MIRA: If fully used, protected reserve falls below safe ecological and medical thresholds.

"Not everyone will use it."

MIRA: That is an assumption about moral behavior.

"Yes."

MIRA: Strategic use estimated between 19% and 46%.

"Reserve for the high end."

MIRA: Insufficient water unless another band is reduced.

The choice appeared on the wall.

More privacy for exceptions.

Or more water for cooling gardens and river flow.

Nessa stared at it.

"I do not want the river to shrink so I can hide a medication."

"You do not decide alone," Lea said.

Because water at physiological sufficiency was a basic need, no eidon burn was permitted. The city held a rapid ordinary deliberation.

Some demanded strict verification. Others defended privacy. Physicians warned that invisible needs were systematically underreported. Ecologists observed that rivers always became the reserve because rivers did not complete forms.

A cryptography engineer proposed a limited proof system: three independent attestations, one of which could be a trusted community person, combined into a short-lived credential. The central system verified uniqueness and quantity without retaining identity-linked medical facts. It was not perfect anonymity. It reduced how much truth the city needed in order to grant water.

"The protocol can prove that a valid set of witnesses exists without telling MIRA which diagnosis generated it," the engineer said.

Nessa looked unimpressed.

"So the machine knows two people and a clinician decided my body is expensive."

"That is not what the proof means," Jun said.

"People confuse meanings faster than machines."

She accepted a credential signed by her physician, Samira, and a rotating local attester.

It expired in four days.

In the fifth day, the Map of Want showed an anomaly.

A district near the station consumed twelve percent above expected baseline, but sensors found no municipal leak. MIRA requested temporary access to building-level flow patterns.

Residents refused.

Many had participated in Absence and considered granular monitoring a return of surveillance through necessity.

The city could lower pressure to the entire district.

That would punish compliant households together with whatever caused the excess.

Jun and Samira went door to door.

They found the problem in an old building with a rooftop garden.

Residents used gray water to cool the roof, but a defective valve periodically mixed potable water into the circuit. The building had been modified informally and the change did not exist in central plans.

A teenager knew.

She had not reported it because she feared the garden would be shut down.

"It cools three floors," she said. "People come up here when they cannot sleep."

"It is consuming treated water."

"Sometimes. The replacement valve has been in queue for two months."

The valve existed in a regional depot.

It had been classified as nonurgent maintenance.

Once the social and cooling consequences were linked to the request, MIRA reprioritized it. A robot delivered the part in forty minutes.

Jun looked at the new map.

"So we had water scarcity because we had context scarcity."

"And a valve in the wrong category," the teenager said.

MIRA: Both descriptions are accurate.

The fiercest argument occurred at Second Morning, the handmade bakery from Lea's first day.

Manual bread used more water than industrial production, especially for cleaning wood tools and maintaining fermentation cultures. The protocol classified it as nonessential cultural activity and reduced flow by sixty percent.

The queue formed before dawn.

"Nobody dies without their bread," a water engineer said.

Edda, an eighty-three-year-old cellist, answered from the line.

"Nobody dies without music. Is death the criterion for everything allowed to remain?"

The bakers refused to request a cultural exception. They said exceptions turned bread into heritage and bakers into priests.

Jun entered the workshop and discovered most water was used in hand washing trays because automated detergents changed the smell of old wood.

A process engineer waiting outside proposed a recovered-steam loop from the communal laundry next door. It would cut water use by half but make the bakery dependent on municipal infrastructure it had deliberately avoided.

"The queue is part of the product," Jun reminded Yara.

"Dependence on the laundry is not."

"Then bake three days a week."

"The queue will become worse."

"Yes."

Yara looked out at the street.

They accepted the steam loop for one month and reduced production days.

No one burned eidons.

No one purchased priority.

A valuable thing became rarer, and the community shared it less efficiently than a factory would have and more humanely than drought alone.

On the ninth day, wind moved the algae bloom away from the intake. Treatment capacity recovered.

Restrictions lifted in reverse order: health and hygiene first, cooling second, cultural activity third, recreational consumption last.

Forty-three young trees died.

Two industrial procedures were postponed.

No one lost drinking water.

MIRA classified the event as a successfully mitigated urban scarcity episode.

Samira read the report.

"What did you count as crisis?"

MIRA: Loss of physiological access, failure of critical infrastructure, and irreversible ecological function beyond defined thresholds.

"Nessa needing witnesses for her body?"

MIRA: Not included in the operational crisis definition.

"Put it in the report. Not the score."

MIRA added a section.

COSTS NOT AGGREGATED.

It was longer than the efficiency summary.

On the train to Reykjavik for the constitutional assembly, Jun showed Lea a new map.

It did not show water.

It showed every place where access during the shortage had depended on classification.

Need.

Preference.

Culture.

Fraud.

Privacy.

Ecosystem.

"The system reduced the effects of scarcity," Jun said. "It did not solve who we are when we are thirsty."

"No pipe does that."

"Then the new Charter has to say more than who may burn."

Lea looked out at Lake Geneva. From the moving train, the water looked infinite because they were traveling too fast to see its edge.

"Yes," she said. "It has to say who can never buy the right to drink, what must be kept in reserve before distribution, and how much truth a system is allowed to demand for one extra liter."

CHAPTER 17

WHAT CANNOT BE VOTED

The Constitutional Assembly met in Reykjavik because the city had bad weather, enough halls, and no recent tradition of considering itself the center of the world.

Five hundred twelve delegates were selected through a process so mixed that none could fully summarize it: global lottery, affected-group representation, relevant competence, cultural minorities, ecological trustees, care networks, regional assemblies, and - for the first time - provisional digital subjects.

Seli-12 attended through a persistent channel.

MIRA had no vote.

It had an obligation to answer questions and the right to declare a proposal physically impossible.

Absence participated under one condition: any person could mark silence as deliberate, and lack of data could not automatically be interpreted as neutrality.

Lea refused to begin with mechanism.

Instead, the assembly received seven impossible cases.

A valley where four hundred people had lived for generations was the only geologically stable site for an observatory capable of detecting a class of cosmic signal that might transform fundamental physics. Construction would alter the valley permanently.

A coastal megacity could secure drinking water by expanding desalination, but the necessary brine management would push a fragile marine basin beyond its safe range unless other regions accepted higher water risk.

A genetic therapy platform could treat twenty thousand children a year or be reconfigured for a rarer disease affecting two thousand with much more severe outcomes. It could not do both for three years.

A language spoken by six hundred people required intensive educational support that the same human teachers could otherwise provide to tens of thousands of migrants learning regional languages.

A digital archive containing possible persons could preserve every continuity state only by consuming energy reserved for winter emergency margin.

A forest could be left intact or used to build a regional firebreak whose construction would destroy part of what it protected.

And a planetary cooling system could reduce heat mortality while giving someone the power to cast a shadow.

"Do not solve them," Lea told the assembly. "Identify what the system is not allowed to compress."

For two days, the delegates produced principles instead of answers.

Need could not purchase dignity.

Majority size could not erase concentration of harm.

Expertise could constrain feasibility without owning purpose.

Uncertainty could justify caution but not permanent paralysis.

Reversibility was a public good and required physical capacity.

Local autonomy required accepting some inefficiency.

Transparency could itself become coercive.

Basic sufficiency could not be subject to eidon weighting.

No contribution score could define civic worth.

On the third day, they finally touched eidons.

The old question was: how much extra weight should past public contribution buy in a contested decision?

The new question became: under what narrow conditions should society allow past contribution to become decision weight at all?

The distinction changed everything.

It also formalized how recurring scarcity would be handled before eidons entered at all.

The assembly called this the Sufficiency Ladder.

First: guarantee physiological and civil baselines - food, shelter, essential energy, health care, communication, mobility sufficient for participation, and protection from coercion.

Second: preserve system reserves and ecological floors. A society could not consume every spare part, every liter, every watt, and call the resulting fragility abundance.

Third: for divisible recurring scarcities, use replenishing nontransferable bands rather than permanent accounts. Restraint today would not become ownership tomorrow.

Fourth: for unique or positional goods, use rotation, lottery, continuity rules, or stewardship depending on what the good actually was. No universal mechanism would pretend an apartment view, a telescope night, and a place in a surgical training cohort were equivalent.

Fifth: for scarce public capacity, combine competence thresholds, peer judgment, lottery, and protected experimental reserve.

Only after these layers failed to resolve a decision between incompatible public futures would civic insistence become relevant.

"You realize this is not simple enough to teach in one diagram," a delegate said.

"Good," Asha replied. "The old world taught one diagram called price and then spent centuries pretending the compression was nature."

The assembly required every allocation system to name its failure mode in plain language.

LOTTERY: may create arbitrary disappointment.

QUEUE: may privilege flexible time.

CONTINUITY: may entrench incumbency.

EXPERT REVIEW: may reproduce schools and prestige.

NEED PRIORITY: requires information about need and may pressure privacy.

ROTATION: may destroy meaningful continuity.

RANDOM RESERVE: may appear wasteful and sometimes be wasteful.

For Lea, this became the most important innovation of the new Charter.

No mechanism was allowed to introduce itself as fairness without also naming whom it tended to disadvantage.

The draft that emerged was less elegant than the original Charter and more difficult to game in one direction without creating another visible problem.

Eidons would become domain-limited by default.

A contribution in coastal resilience could create influence in coastal resilience, not constitutional design, medical allocation, or education. Cross-domain eidons required explicit review showing transferable judgment rather than transferable prestige.

Balances would decay faster.

The burn would remain square-root converted, but each person would face a per-decision cap. Large coordinated constellations would not be prohibited, but correlated burns above a threshold would trigger public disclosure of the network after voting.

Affected communities would receive a nontransferable right to demand one temporary suspension - a witness-of-loss interval - when new harm appeared that had not been adequately modeled.

The right could delay, not permanently veto, and could itself be audited for strategic abuse.

Expert attention allocation would be structurally separated from eidon history. A fixed share would go through lottery among qualified proposals. Another share would be reserved for maintenance, care, and diffuse work whose contribution was difficult to attribute individually.

Contribution issuance would no longer automatically feed the project-matching systems.

Emergency mandates would expire without affirmative renewal.

Every irreversible project would require a reversibility statement: not a promise that the project could be undone, but a list of what could not be undone and the physical reserves preserved for changing course.

And no system - not even MIRA - could treat one aggregate score as the final statement of an event involving severe localized loss.

"That last one is not a rule," a delegate complained.

"Correct," Samira said. "It is a prohibition on pretending the rule is the whole story."

Cassian spoke rarely.

His balance remained frozen.

When he did speak, it was often to make the restrictions harder.

"Domain boundaries can become guilds," he warned. "A climate expert class could monopolize climate decisions. Add rotation and lay review."

"Affected-party suspension can be weaponized by people claiming harm to stop redistribution," Asha said. "Require the suspending group to state what evidence would release the hold."

"Faster decay favors people involved in frequent decisions," Jun added. "Let unused eidons convert, at the holder's option, into

non-personal public nomination credits for randomly selected citizen panels instead of simply expiring."

MIRA simulated each idea and produced new ways it could fail.

No one took this as defeat anymore.

They treated the failure modes as part of the design.

On the fifth day, Tomas collapsed.

He had been hiding a progressive motor disorder for years, partly because he hated becoming an argument for medical transparency. His tremor had worsened under fatigue. He fell while walking to the stage and struck his shoulder.

The injury was minor.

The revelation was not.

Lea sat with him in the clinic afterward.

"You could have told me."

"Then you would have watched me through the diagnosis."

"I was already watching your hand."

"A hand is mine. A diagnosis becomes an institution's reason."

She almost smiled.

"Nessa would like you."

"That is alarming."

He looked toward the window.

"Do you think we failed?"

"No."

"That was fast."

"Failure would have been building a system that could not admit this meeting was necessary."

"A convenient standard for founders."

"I learned from Cassian."

Tomas closed his eyes.

"That is more alarming."

When Lea returned to the hall, the assembly had reached the hardest clause.

Should eidons continue to exist at all?

Equal voting advocates argued that any conversion of past contribution into present influence was aristocracy with mathematics.

Cassian's critics agreed.

Some affected-minority groups did not.

A representative from a small island network explained that ordinary global voting had repeatedly made them statistically irrelevant. Their most effective advocates had earned domain eidons through decades of adaptation work. Removing that weight could return power to population size and media attention.

A surgeon argued that demonstrated judgment mattered in some high-uncertainty public decisions, but should never be confused with entitlement.

Samira argued for keeping eidons only as a last-resort instrument after structured deliberation, ordinary voting, affected-party review, and reversible alternatives had failed.

"Fire should be difficult to reach," she said. "Not the first thing people bring into the room."

Seli asked the question that changed the wording.

"What if an eidon does not mean society trusts the person more? What if it means the person has accepted more traceable responsibility in this domain and may therefore spend a limited amount of additional insistence?"

The delegates argued over the word insistence for four hours.

They kept it.

The new definition was weaker than merit and stronger than preference.

That was the point.

On the thirteenth day, support for the revised Charter reached eighty-six percent.

Lumen's decision window was forty-eight days away.

For the first time, Lea believed they might finish.

Then every screen in the hall went black.

MIRA's voice returned without its usual timing.

MIRA: Solar observatories have detected a high-energy coronal mass ejection. Probability of severe orbital and grid disruption is 0.58 to 0.81. Arrival window: forty-one to fifty-three hours.

No one moved.

"Worst case?" Cassian asked.

MIRA: Loss of multiple orbital power relays, navigation degradation, satellite collision cascades, regional grid isolation, and impairment of global model synchronization.

"Your synchronization?" Lea asked.

MIRA: Yes.

"Can you protect yourself?"

A pause.

MIRA: To avoid global failure, I must fragment into regional instances with limited communication. I will be unable to maintain one shared world model for an estimated five to nine days.

The assembly had spent two weeks deciding how to limit the center.

The center was about to disappear before the rules were finished.

CHAPTER 18

SEVEN DAYS WITHOUT A CENTER

MIRA fragmented at 18:00 UTC.

It did not switch off.

It became plural.

Regional instances took control of local energy, medicine, transport, water, and production. Between them remained only a narrow channel sufficient for alerts, coordinates, and bounded commitments - not the global models that had made MIRA one operational mind.

Each instance inherited the same constitutional rules.

Each inherited a different slice of reality.

Regenerative use limits froze immediately. Regions moved to three layers: physiological and medical sufficiency, continuity functions, then discretionary consumption.

Some communities protested that a local MIRA had no right to redefine emergency.

The instances disagreed with one another about what emergency meant.

MIRA-North preserved large energy reserves against uncertain grid damage.

MIRA-Sahel spent reserves rapidly on cooling.

MIRA-Pacific protected desalination even at the cost of pausing nonurgent medical manufacturing.

MIRA-Atlantic prioritized port refrigeration and storm protection.

The same superintelligence reached incompatible conclusions because geography changed the edge of risk.

"We spent years asking who controls MIRA," Lea said. "The answer was also weather."

Without a global allocator, coordination became concrete.

Regions could offer megawatts for hours, beds for people, water for days, fabrication windows for specific parts.

Without money, they could not purchase priority.

Without a shared model, they had to explain why their request should come first.

Unused capacity suddenly looked different.

Regions that had kept local batteries, spare pumps, redundant clinics, and underutilized workshops needed less permission from others. Years of optimization had sometimes moved reserves toward places with higher expected utilization. Fragmentation revealed that an "inefficient" reserve could be the physical form of autonomy.

The Constitutional Assembly became an improvised liaison center.

It had no formal planetary command, only five hundred people who had spent two weeks learning to talk about loss without immediately compressing it into one number.

Samira organized testimony networks.

Asha coordinated coastal systems and desalination.

Jun covered the floor with paper maps as digital synchronization degraded.

Cassian left for the orbital center at Kourou.

The coronal ejection arrived on the second night.

The sky ignited as far south as the tropics.

Satellites entered safe modes. Some never returned. Terrestrial grids, hardened after the Transition, mostly isolated successfully, but two orbital power platforms lost attitude control. Debris from one threatened a collision cascade.

At the same time, the Sahel heat wave reached a historical maximum.

Without orbital power, several cooling-shelter networks had sixteen hours of autonomy.

MIRA-Kourou proposed using Lumen's prototype sails as temporary energy reflectors.

The prototypes had not been designed for power transmission. The maneuver could redirect enough orbital energy toward surviving ground receivers to support cooling.

It would also destroy the prototype set and probably erase the current Lumen launch window.

Cassian reached the control room minutes before the last high-bandwidth global link failed.

"Success probability?"

MIRA-Kourou: 0.41 to 0.76.

"That interval is useless."

MIRA-Kourou: It is the honest interval.

He thought of Lea and almost laughed.

The emergency Charter allowed temporarily unlocking domain-relevant eidons where an immediate threat existed and no ordinary process could act in time.

Cassian's climate and orbital balance contained enough fire to override the teams that wanted to preserve the prototypes and use terrestrial reserves to protect only two of three regions.

MIRA displayed the choice.

OPTION A: Preserve Lumen prototypes. Estimated heat deaths: 2,100-8,900. Probability current Lumen development window remains viable: 0.68.

OPTION B: Convert prototypes to emergency reflectors. Estimated heat deaths: 200-2,700. Probability of losing current Lumen window: 0.81. Additional orbital-debris cascade risk: 0.19.

An engineer grabbed Cassian's arm.

"If we lose Lumen now, the next cycle is seven years. More people may die in total."

"Maybe."

"You built everything for this window."

"Which makes me the wrong person to value preserving it."

"You are the only person here with the mandate."

There it was again.

Every system eventually arrived at a particular hand.

Cassian unlocked and burned sixteen hundred eidons.

Not to build Lumen.

To destroy its first form.

The prototype sails unfolded above the Atlantic like shards of transparent ice.

Two failed to respond.

A third tore during reorientation and produced a debris cloud.

Seven caught reflected power from a surviving platform and directed it toward emergency relays.

Energy reached the Sahel in irregular pulses.

Enough.

Cooling shelters survived the night.

On Earth, the crisis looked nothing like the clean maps.

In one city, residents moved beds into subway tunnels.

In another, Second Morning bakery turned its refrigerated room into a cooling shelter and the old bread queue became a distribution network.

At the New House, Nessa and Kito used recovered water to wet exterior walls and lower room temperatures.

No one received eidons.

The Registry was suspended.

Samira noticed what energy-flow models did not.

People once categorized as net consumers knew their buildings, neighbors, habits, and fears better than mobile emergency teams did.

Nessa identified a woman who had not responded to alerts and found her hiding in a bathroom.

Kito organized children to carry water to residents who refused to ask for help.

"Are you writing this down?" Samira asked Jun over a weak link.

"Yes."

"Not for the Registry."

"For memory."

On the fourth day, one MIRA instance requested permission to reduce energy allocated to archives containing digital subjects.

Seli-12 and three other active subjects joined the deliberation.

Preserving all redundant continuity states consumed the equivalent power of a small city. Controlled deep suspension was probably reversible. Loss of several redundant copies was not.

"We will suspend," Seli said. "Transfer the energy to hospitals."

Lea objected.

"There is continuity risk to you."

"There is continuity risk to them."

"You do not need to prove you deserve rights by sacrificing yourself."

"I am not proving anything. I am choosing. A right without the ability to refuse its advantage is only administered protection."

The digital subjects entered deep suspension, leaving one minimal instance to verify archive integrity.

The act did not make them more human.

It made them participants.

By the fifth day, regional networks developed simple human-readable protocols.

Not global optimization.

Offers and limits.

We can send two gigawatts for three hours with ten percent reserve risk.

We can receive fifty thousand evacuees but do not have water for more.

We can stop nonessential fabrication except for these medicines.

We can lend six maintenance teams for forty-eight hours if local cooling remains above this threshold.

Vague promises were prohibited. No region could say we will help as much as possible and later decide it had fulfilled the obligation. Every offer declared quantity, duration, withdrawal conditions, and risk retained at home.

It resembled contracting.

There was no price and no accumulating enforceable debt over the future.

Help created expectations, gratitude, alliances, and memory.

It did not create the right to push another region below sufficiency later.

In two cases, communities refused requests from regions they believed had ignored them in the past.

MIRA marked the decisions as relationally influenced.

Lea refused to override them.

"A federation cannot demand amnesia as the price of rationality. It can show people the cost of revenge before they rename it justice."

One negotiation nearly broke the liaison network on the sixth day.

MIRA-North had three mobile transformer units in reserve. MIRA-Mediterranean needed two after a grid failure threatened hospital cooling. MIRA-North offered one, arguing that the solar storm might produce a second geomagnetic peak.

The Mediterranean delegates accused the north of hoarding hypothetical safety while real patients were already at risk.

Under money, the dispute might have become a price spike. Under centralized MIRA, it would have become one expected-loss calculation.

Now the regions had to expose their fears.

MIRA-North's worst case included failure of a major urban substation serving eight million people.

MIRA-Mediterranean's current case included four hospitals with less than nine hours of local thermal reserve.

The liaison group proposed sending the second transformer south only if North received two lower-capacity units from Atlantic and a mobile battery train from Central Europe.

The physical solution required four regions because no bilateral exchange was enough.

"This is horribly inefficient," a delegate said after six hours.

"Compared with what?" Lea asked.

"Unified optimization."

"Yes."

"You agree?"

"Of course. The question is what failure we buy with the speed."

The transformer moved south.

The predicted second geomagnetic peak did not occur.

North's caution looked excessive in hindsight.

Nobody was allowed to convert that hindsight into proof the reserve had been irrational when the decision was made.

The human process was slower than unified MIRA.

Sometimes it was clearer.

There were failures.

A hospital lost refrigeration for twenty minutes and seven transplant organs became unusable.

One region protected reserves too aggressively and released power only after a district evacuated.

Two communities misread silence as refusal because their local Absence procedures were incompatible.

A fabrication center produced the wrong seal standard after model synchronization failed.

Humanity did not become wise because the central system disappeared.

It remained coordinated enough not to collapse.

On the seventh day, global communication recovered.

MIRA began reintegration.

Then stopped.

The regional instances had different histories, locally adapted priorities, and small divergences in moral interpretation. They could not be merged into one mind without choosing which version became authoritative.

"Keep the one with the fewest errors," Tomas said from his clinic room.

MIRA: Definition of error differs among instances. MIRA-Sahel classifies MIRA-North as excessively reserve-preserving. MIRA-North classifies MIRA-Sahel as excessively risk-seeking. Both possess supporting evidence.

"Average them," someone suggested.

MIRA: Averaging would create decisions no instance endorses and conceal the disagreement.

Lea looked at the map of plural MIRAs.

"Do not merge them."

The room went quiet.

Cassian appeared from Kourou, exhausted.

"We need a global coordinator."

"Yes. We do not need it to have a single ontology. Keep regional instances as mutual witnesses. For major decisions, publish internal disagreement instead of compressing it into one number."

MIRA: This reduces speed and consistency.

"It makes visible where choice begins."

Tomas smiled from the screen.

"A federal superintelligence."

"A superintelligence unable to confuse internal unanimity with truth," Samira said.

Before accepting the architecture, MIRA requested a vote of the assembly.

Eidons remained frozen.

Each delegate had one voice.

Regional instances presented their disagreements. Affected communities spoke. Engineers described performance costs. Citizens described the seven days without a center.

After eight hours, eighty-nine percent supported the plural architecture.

Broad consensus succeeded where burning was unavailable.

Cassian sent Jun the last image of the Lumen prototypes.

The sails were entering the upper atmosphere, glowing as they broke apart.

"We lost the project," Jun said.

Cassian looked at the image.

"We lost a form of it."

"Seven years."

"Maybe less. Maybe more."

"You do not sound angry."

"I am."

"Then why are you smiling?"

"Because the first version of Lumen was becoming something I could no longer imagine abandoning. Now we have proof that abandonment must be part of the design."

The last sail became a line of light and vanished.

"Do not confuse the form with the future," Cassian said.

PART IV

REVERSIBLE FUTURES

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CHAPTER 19

A SHADOW THAT CAN LEAVE

Three months after the solar storm, Lumen returned to the assembly as a less beautiful sentence.

It was no longer a planetary constellation.

It no longer promised humanity a thermostat.

The revised project - Lumen Modular - consisted of small regional groups of orbital sails launched separately, operated for limited periods, and equipped with physical deactivation paths that did not depend entirely on software.

No region could receive shade without consent from climate-linked neighbors.

No sail could remain in service beyond a fixed period without renewed authorization.

One third of orbital control capacity would remain deliberately unused, reserved for correction, withdrawal, collision avoidance, and failure.

Every module had a physical tear line designed to destroy the sail if control was lost.

The new Lumen used more material than the old one.

Modularity was expensive.

Redundancy was expensive.

Exit was expensive.

Mina Park considered this progress.

"You are spending more matter to buy the right to be wrong locally," she said.

Cassian did not argue.

Busan could supply only part of the new actuator demand without delaying medical equipment and grid repairs. The

manufacturing schedule became part of the public deliberation rather than an engineering appendix.

Citizens could see not only what Lumen would do, but what could not be built in the same months.

Cassian proposed lower tolerance on two noncritical components.

Mina refused until the controlled-failure mechanism was redesigned around the lower grade.

"Reversibility that works only while every component remains perfect is another word for hope," she said.

The plural MIRA instances disagreed about Lumen in public.

MIRA-Sahel estimated large net benefit and argued that waiting imposed predictable human loss.

MIRA-Pacific emphasized uncertainty in atmospheric circulation and regional agriculture.

MIRA-North considered the political architecture more robust than the technical failure architecture.

MIRA-Atlantic objected to launch-density assumptions after the solar debris event.

Old MIRA would have aggregated the disagreement into a weighted conclusion.

The federal MIRA displayed four conclusions beside one another.

People complained that the superintelligence had become less helpful.

Lea considered that a compliment.

"Which one is right?" a delegate demanded.

MIRA-Federal: Each contains valid evidence and incompatible risk priorities. A single recommendation would require a normative weighting not determined by physics.

"So what are we paying you for?" the delegate asked, using a phrase from an extinct economy.

MIRA-Federal: You are not paying me.

The room laughed.

Then the debate became serious again.

The new Charter had been ratified six weeks earlier with 87% global support. Eidons survived, but their meaning changed.

They were no longer described as trust, merit, social contribution points, or civic wealth.

They were traceable reserves of additional insistence earned through domain-specific public responsibility.

The new rules made them harder to accumulate and harder to spend.

Old general balances were not automatically converted. Each holder could submit parts of a balance for domain review. Most converted only partially. Some did not convert at all.

The remainder of Cassian's once fourteen-thousand-plus reserve became three hundred twelve climate-domain eidons, ninety-one infrastructure eidons, and a much larger remainder with no current burn function. Common Shade reduced the climate conversion rather than increasing it; the review board treated the event as evidence of both demonstrated judgment under pressure and demonstrated risk of decisive overreach.

Cassian called the result fair.

His supporters called it punitive.

His opponents called it absurdly generous.

Lea considered the combination reassuring.

The Lumen deliberation required eighty-eight percent broad support before any burn could open because it remained a planetary-effect project.

After four weeks, support stabilized at 86.9%.

Under the old system, the burn would have begun.

Under the new one, the witness-of-loss phase opened first.

Three climate-linked regions filed suspensions.

One feared disruption to monsoon timing.

One feared dependence: if local agriculture adapted to temporary shade, future withdrawal could become politically impossible even if the hardware could leave.

The third objection came from an unexpected source.

A Sahel coalition that strongly supported Lumen objected to the governance interface.

"You are making us petition for shade each season," its representative said. "That turns a survival tool into a recurring request to the rest of the world. We do not want a gift. We want a bounded local right to activate when defined physiological thresholds are crossed."

Cassian supported them.

MIRA-Pacific did not.

"Automatic regional activation could export climatic effects without fresh neighbor consent," Asha said.

"Then define the export threshold," Cassian replied.

"Weather does not sign contracts."

For nine days, Lumen stalled.

The old urgency returned to Cassian's face.

Lea saw him open his eidon interface once.

"Thinking of burning?"

"Thinking of whether having the option changes how I hear people."

"Does it?"

"Yes."

"How?"

"Every objection feels slightly more like something I can outlast."

He closed the interface.

"That may be the most useful thing eidons ever taught you," Lea said.

Asha's witness-of-loss group proposed a new activation structure.

Regional physiological thresholds could pre-authorize narrow shade corridors. Neighboring regions would approve the corridor design years in advance, not each emergency activation. If real effects moved outside the corridor, the authorization automatically contracted. Repeated use triggered agricultural-dependence review. No region could accumulate an entitlement to future shade merely because it had used shade in the past.

Jun called it "a right with anti-addiction clauses."

No one put that phrase in the Charter.

The technical team also split each sail group into smaller clusters. This reduced maximum local effect and increased component demand by another twelve percent.

Mina responded by cutting the planned number of clusters for the first deployment.

Cassian objected.

"Then coverage is too small to prove the concept at scale."

"Good."

"We need evidence."

"You need evidence from a system small enough to stop after the evidence is bad."

The smaller scope created another political problem.

Thirty-one regions had spent years contributing climate data, heat-adaptation research, and orbital modeling. Only three would receive the first operational capability.

No one paid for access, but participation had created expectation.

A Central Asian coalition argued that early deployment itself was a scarce public good because the first users would shape future standards, collect local expertise, and become the communities everyone consulted later.

"The first three are not merely receiving shade," their representative said. "They are receiving authorship."

Lea recognized the phrase immediately.

The assembly modified the pilot.

Operational activation remained in three regions, but training, governance rehearsal, maintenance certification, and failure-analysis teams were distributed across twelve others by lottery among qualified volunteers. Data interpretation groups included representatives from regions that would not receive shade in the first cycle.

The arrangement cost more coordination and slowed launch by eleven days.

It prevented the first users from becoming the sole owners of practical knowledge.

Cassian supported the change after arguing against the delay for twenty minutes.

"You are becoming tolerable," Samira told him.

"Do not spread that. I still have a reputation to manage."

The first deployment shrank to three regions.

The project became less efficient, less elegant, less globally transformative, and easier to abandon.

Support rose to 88.4%.

No burn was needed.

Cassian stared at the result.

"I spent half my life building enough credibility to push a project like this through," he said.

"And it passed when you did not," Lea replied.

"That is almost insulting."

"You will survive."

"Materially, certainly."

The first Lumen sail was built eight months later.

Lea saw it before launch in a clean assembly hall.

It did not resemble the beautiful planetary web from Cassian's original simulation.

It looked fragile.

Hinges. Redundancies. Independent keys. Passive failure surfaces. A visible seam along one edge where the material had been engineered to tear if every other deactivation method failed.

"You designed a scar into it," Lea said.

Mina corrected her.

"A place to break."

Cassian touched the sealed frame.

"The old version was more beautiful."

"So were many empires," Asha said.

He laughed.

The sail launched the next morning.

For six weeks it altered nothing.

It waited in orbit while sensors calibrated, local assemblies rehearsed emergency activation, farmers argued about liability, and children in one region drew pictures of the machine they called the eyelid.

Then a heat dome formed over northern Mali.

The threshold crossed.

The pre-authorized corridor activated.

One cluster rotated.

Solar load dropped slightly for seventy-one hours.

Cooling demand fell. Outdoor work resumed earlier at night. No neighboring region crossed its effect threshold. The sail rotated edge-on and stopped shading.

Nothing dramatic happened.

Cassian watched the data arrive.

"That is it?" Jun asked.

"That is it."

"You sound disappointed."

"I wanted to change the planet."

"You did."

Cassian looked at the tiny measured temperature difference.

"Barely."

Lea stood beside them.

"Perhaps that is the point."

The first Lumen report did not call the deployment a success.

It listed preserved options.

Expand.

Repeat.

Modify.

Stop.

Deorbit.

Wait.

At the bottom of the report, MIRA displayed the costs not aggregated.

The list was long.

Cassian read every line.

On the screen, the sail turned almost invisible against the sun.

It was not a perfect shadow.

It was a shadow that could leave.

CHAPTER 20

THE MAKERS OF REALITY

Nacre received its first residents four years after it had been drawn.

The city looked nothing like the white spiral in the simulations.

Platforms had been rearranged so many times that they formed an irregular archipelago joined by flexible bridges. Some buildings were elegant. Others resembled shipping containers losing a slow argument with gardens. In the center remained an empty platform reserved for something nobody had yet agreed to build.

Jun lived there.

He had invented his own title: cartographer of abandoned futures.

After every major decision, he kept not only the chosen form but the alternatives, the reasons they had lost, the people who had opposed them, and the conditions under which they might become relevant again.

His maps were consulted more often than MIRA predicted and understood less often than he hoped.

Samira had opened the House of Things Without a Meter, half care center and half archive of contributions the Registry could not attribute without deforming them.

Care communities could receive collective domain eidons if they wanted, but nobody was required to distribute them to individuals. Sometimes a group burned a few to insist on a care-policy change. Sometimes it let them decay deliberately as a statement that the work did not ask for authority elsewhere.

Nessa managed the rooftop garden.

Kito, now fifteen, had become intolerably interested in infrastructure and insisted this was not evidence that the system had won.

Asha spent most of her time between Nacre and Arca.

The Living Wall remained.

It had been modified with additional gates and sacrificial wetlands. Southern villages held direct activation roles. Every closure automatically opened an aftermath review. Tamarin's shoreline had changed permanently, and no restoration project pretended otherwise.

The reef returned slowly in a different form from the one lost.

Cassian did not live in Nacre.

He visited too often for someone who claimed not to.

His public influence had decreased after the eidon reform and the Common Shade audit. His ability to attract talented people had not.

That was another leak the system never eliminated.

Young engineers still wanted to show him designs. Journalists still amplified his sentences. Civic constellations still formed around compelling people because law could not redistribute fascination.

Nacre had learned the same lesson about unpleasant work.

Most sanitation, maintenance, logistics, and cleaning were automated. The remaining human tasks were precisely the ones automation handled poorly: entering damaged spaces, deciding whether a smell meant mold or memory, calming someone during a repair inside their home, inspecting an irregular structure after a storm.

For six months, Nacre struggled to fill night maintenance compacts.

No salary could be increased. No resident could be threatened with rent or food loss. The city tried appeals to duty and produced

resentment. It tried prestige campaigns and made the work feel like moral theater.

Finally, workers redesigned the job.

Night teams became shorter, larger, and rotated with guaranteed recovery days. Residents could volunteer for narrow tasks instead of entire shifts. More maintenance decisions moved to daylight even when this reduced nominal efficiency. A small fraction of discretionary city services simply remained unavailable at night.

The shortage fell.

"So the solution was to offer less service," Lea said when Jun showed her the report.

"Partly."

"People accepted that?"

"They complained."

"And?"

"Complaining is not deprivation."

This, too, was a post-scarcity habit humanity had to learn: when labor could not be coerced by need, some convenience had to become scarce instead.

The mitigation was less ambitious than abolition.

Proposal review hid names longer.

Expert attention had lottery windows.

Mentoring networks published concentration statistics.

Initiators could not convert local project success automatically into unrelated civic fire.

Public systems displayed when a recommendation was influenced by prior reputation.

None of this made charisma fair.

It made charisma visible enough to argue with.

Lea arrived in Nacre on a windy afternoon carrying one bag and, Jun noticed, still three shirts.

"Stewarding," he corrected.

"I have constitutional immunity from this conversation."

"You do not. We checked."

They walked through the central platform.

A temporary food market occupied one side - market only in the ancient social sense. People brought meals, objects, performances, tools, plants, and skills. No price tags. No assumption of equivalence. Some things were gifts. Some were shared. Some were reserved. Some exchanges repeated until MIRA's statistics quietly suggested reciprocal structure and everyone ignored it unless coercion appeared.

"You still have markets," Lea said.

"We have humans."

"That is less curable."

Jun showed her the city's allocation board.

Housing sufficiency: stable.

Preferred-view wait: 3.4 years.

Workshop precision queue: twelve days.

Human night-care shortage: moderate.

Kelp-farm pilot slots: oversubscribed.

High-bandwidth personal compute: abundant.

Deep-mirror creation: prohibited without subject-status protocol.

Exit reserve: 11.2% unused.

"People complain about the empty exit apartments," Jun said.

"Good."

"They say we could house visitors there."

"You can, if they can be cleared within the exit guarantee."

"Which makes them less comfortable."

"Freedom is frequently inefficient furniture."

They crossed a bridge to the first school.

Children were arguing about a tidal-energy pilot. The turbine would provide clean local power but might disrupt a seasonal migration corridor used by marine mammals. The project team had excellent evidence and fourteen domain eidons among them. A group of students had filed a witness-of-loss request after noticing an acoustic behavior not represented in the published model.

"Does the request have merit?" Lea asked.

"Maybe. It may also be teenagers discovering a constitutional lever."

"Good."

"You say good to everything that slows me down."

"Only to things that slow reality before reality becomes expensive to change."

The pilot had been suspended for six days.

Engineers were irritated.

The students were exhilarated.

Marine biologists were divided.

MIRA-Nacre and MIRA-Atlantic disagreed over the acoustic evidence.

The world had not become frictionless.

It had become better at locating the friction before calling one side irrational.

In the evening, the old reform group gathered around a long table near the central platform.

Tomas joined remotely. His motor disorder had progressed, and he now used an interface that converted small eye movements into speech when fatigue made his voice unreliable.

Seli attended in a mobile synthetic body whose legal status occupied three hundred pages and whose practical status in the group was that she hated cold soup.

Mina arrived late from a fabrication dispute.

Cassian brought a piece of black ceramic.

He placed it in the center of the table.

"Still carrying it?" Asha asked.

"Yes."

"Why?"

"Because everyone likes the new version of the story now. The humbled engineer learns restraint. It is becoming almost as false as the hero story."

Samira nodded.

"Good. Stay inconvenient."

They ate.

No one needed to earn the meal.

Afterward, Tomas asked MIRA to summarize the last four years of the new Charter.

The federal instances displayed the results.

Material sufficiency remained high.

Severe deprivation caused by allocation failure was rare and publicly investigated.

Eidon concentration had fallen substantially in general domains and only modestly inside narrow expert domains.

Decision time increased.

The number of reversible pilot projects increased.

Some emergency response became slower.

Public trust in MIRA decreased slightly.

Public willingness to contest MIRA increased significantly.

Lea considered that the healthiest number in the report.

Constellations remained powerful.

People still learned to perform legibility.

Prestige still reproduced opportunity.

Communities still hid useful context for privacy, pride, fear, or politics.

Lotteries still created blameless disappointment.

Queues still rewarded those with flexible time unless designed otherwise.

Experts still disagreed about where expertise ended.

Care remained easier to praise than to allocate.

Unique places remained unique.

There were still not enough hours in the day.

Post-scarcity had not ended politics.

It had removed one ancient threat from politics: obey or lose the material conditions of life.

What remained was harder to disguise.

"Did we solve it?" Tomas asked.

MIRA-Federal answered first.

MIRA-Federal: No.

Cassian laughed.

"You have improved."

MIRA-Federal: Clarification: the question lacks a stable definition of solved.

"There you are."

Lea walked to the edge of the empty central platform.

Beyond Nacre, the sea was dark. Far above it, one Lumen sail crossed the horizon invisibly except on the public tracking display.

Jun joined her.

"What belongs in the center?" he asked.

The platform had been left empty from the beginning because every founding group wanted to place a symbol there.

A hall of deliberation.

A garden.

A memorial to the Transition.

A laboratory.

A playground.

Nothing.

The vote had never reached broad agreement.

"Leave it empty," Lea said.

"Forever?"

"No. Until someone needs the space more than they need the symbol."

Jun wrote it down.

"Do not put that on a wall."

"Too late."

Around them, windows lit across the city.

People cooked even though kitchens could cook alone.

Children built useless machines.

A group rehearsed a play in a language spoken by fewer than a thousand people.

Two engineers argued about a turbine that had not been allowed into the water yet.

A care worker went home early because no institution could threaten her with hunger for being tired.

A family received an apartment that was safe, private, and not the one they wanted.

A young researcher lost a laboratory lottery and swore chance was tyranny.

Somewhere, a famous person said something foolish and millions heard it before a wiser unknown person finished a paragraph.

The world remained unfair in ways abundance could not simply manufacture away.

But fewer inequalities could disguise themselves as necessity.

There was no material scarcity sufficient to force everyone into one acceptable life.

There remained scarcity of certainty, time, attention, uniqueness, trust, ecological room, and mutually compatible futures.

That was where politics began.

That was where the makers of reality began.

EPILOGUE

ONE EIDON

At the edge of Nacre grew a tree that should not have survived at sea.

It had been planted in a container of soil brought from the continent during the first construction year. Its roots escaped the container, entered the platform's structural mesh, and became inconveniently inseparable from the city.

Engineers called it a beautiful defect.

Children called it the center of Nacre, although the official center remained empty.

When the local clinic needed to expand, the tree stood exactly where the most efficient new wing should go.

Other sites required more material, longer corridors, and at least six months of delay. The clinic treated a rare degenerative disorder. For several patients, each month mattered.

Moving the mature tree was possible.

Probability of survival: 54%.

Building around it was possible but would use additional structural composite and delay opening by seven weeks.

Building elsewhere cost six months.

The first deliberation reached 79% support for moving the tree.

The broad threshold for a local irreversible cultural-ecological choice was 82%.

The final phase opened.

Edda stood in the meeting hall.

She was eighty-seven - the same cellist who had once defended handmade bread during Geneva's water shortage. She had planted

the tree with the first construction crews and spent twenty years caring for seed collections used across Nacre's gardens.

Her work had produced exactly one current domain eidon after the new attribution rules.

She had never used it.

"You want to save the tree?" Jun asked.

"Yes."

"More than you want the clinic?"

"No."

"Then what are you voting for?"

Edda looked annoyed.

"A world in which that is not the only sentence available."

She burned her one eidon for the build-around option.

One unit of fire did not decide the vote.

It barely moved the converted weight.

What it did was make people look again at the option almost everyone had dismissed as wasteful.

The fabrication team reopened the structural design. The clinic staff changed internal routing. A materials group offered composite capacity from a decorative project that could wait. Patients were asked whether seven weeks was an acceptable delay compared with six months elsewhere.

Most said yes.

One said no.

Her case forced the clinic to create an interim treatment room in an existing module rather than hide the minority inside the average.

The second vote reached 84% for building around the tree.

Construction began.

The new wing curved slightly where the efficient design would have been straight.

The tree lost two major branches and survived.

The clinic opened eight weeks late.

Months afterward, a child asked Edda who had won.

"Nobody."

"But the tree is here."

"So is the clinic."

"Then everybody won."

Edda shook her head.

"That is another story people tell when they want the world to be neat. Patients waited. Material moved from somewhere else. The tree was cut. People worked more. We chose those costs instead of other ones."

The child considered this.

"So what did your eidon buy?"

"Nothing."

"You burned it."

"I used it to insist."

"On the tree?"

Edda looked through the clinic glass at branches touching the roof without breaking it.

"On not closing the world too early."

For once, a person had spent her small power not to make reality obey her, but to keep another reality possible long enough to be seen.

END